

APPLICATIONS OF DERIVATIVES

In Section [2.2](#) we defined the derivative at $x = a$, $f'(a)$, of an abstract function $f(x)$, to be its instantaneous rate of change at $x = a$:

$$f'(a) = \lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}$$

This abstract definition, and the whole theory that we have developed to deal with it, turns out to be extremely useful simply because “instantaneous rate of change” appears in a huge number of settings. Here are a few examples.

- If you are moving along a line and $x(t)$ is your position on the line at time t , then your rate of change of position, $x'(t)$, is your velocity. If, instead, $v(t)$ is your velocity at time t , then your rate of change of velocity, $v'(t)$, is your acceleration. We shall explore this further in Section [3.1](#).
- If $P(t)$ is the size of some population (say the number of humans on the earth) at time t , then $P'(t)$ is the rate at which the size of that population is changing. It is called the net birth rate. We shall explore it further in Section [3.3.3](#).
- Radiocarbon dating, a procedure used to determine the age of, for example, archaeological materials, is based on an understanding of the rate at which an unstable isotope of carbon decays. We shall look at this procedure in Section [3.3.1](#).
- A capacitor is an electrical component that is used to repeatedly store and release electrical charge (say electrons) in an electronic circuit. If $Q(t)$ is the charge on a capacitor at time t , then $Q'(t)$ is the instantaneous rate at which charge is flowing into the capacitor. That's called the current. The standard unit of charge is the coulomb. One coulomb is the magnitude of the charge of approximately 6.241×10^{18} electrons. The standard unit for current is the amp. One amp represents one coulomb per second.

3.1 ▲ Velocity and Acceleration

If you are moving along the x -axis and your position at time t is $x(t)$, then your velocity at time t is $v(t) = x'(t)$ and your acceleration at time t is $a(t) = v'(t) = x''(t)$. Your speed, i.e. the distance that you are travelling per unit time, is $|v(t)| = |x'(t)|$.

Example 3.1.1

Suppose that you are moving along the x -axis and that at time t your position is given by

$$x(t) = t^3 - 3t + 2.$$

We're going to try and get a good picture of what your motion is like. We can learn quite a bit just by looking at the sign of the velocity $v(t) = x'(t)$ at each time t .

- If $x'(t) > 0$, then at that instant x is increasing, i.e. you are moving to the right.
- If $x'(t) = 0$, then at that instant you are not moving at all.
- If $x'(t) < 0$, then at that instant x is decreasing, i.e. you are moving to the left.

From the given formula for $x(t)$ it is straight forward to work out the velocity

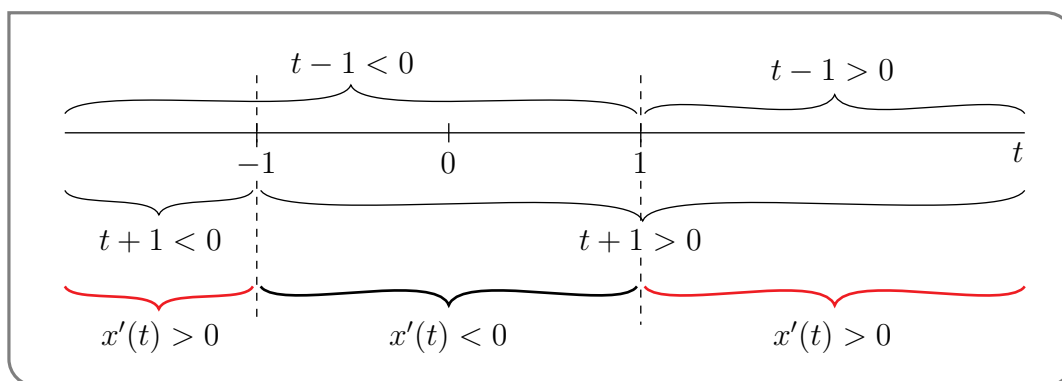
$$v(t) = x'(t) = 3t^2 - 3 = 3(t^2 - 1) = 3(t + 1)(t - 1)$$

This is zero only when $t = -1$ and when $t = +1$; at no other value¹ of t can this polynomial be equal zero. Consequently in any time interval that does *not* include either $t = -1$ or $t = +1$, $v(t)$ takes only a single sign². So

- For all $t < -1$, both $(t + 1)$ and $(t - 1)$ are negative (sub in, for example, $t = -10$) so the product $v(t) = x'(t) = 3(t + 1)(t - 1) > 0$.
- For all $-1 < t < 1$, the factor $(t + 1) > 0$ and the factor $(t - 1) < 0$ (sub in, for example, $t = 0$) so the product $v(t) = x'(t) = 3(t + 1)(t - 1) < 0$.
- For all $t > 1$, both $(t + 1)$ and $(t - 1)$ are positive (sub in, for example, $t = +10$) so the product $v(t) = x'(t) = 3(t + 1)(t - 1) > 0$.

The figure below gives a summary of the sign information we have about $t - 1$, $t + 1$ and $x'(t)$.

- 1 This is because the equation $ab = 0$ is only satisfied for real numbers a and b when either $a = 0$ or $b = 0$ or both $a = b = 0$. Hence if a polynomial is the product of two (or more) factors, then it is only zero when at least one of those factors is zero. There are more complicated mathematical environments in which you have what are called “zero divisors” but they are beyond the scope of this course.
- 2 This is because if $v(t_a) < 0$ and $v(t_b) > 0$ then, by the intermediate value theorem, the continuous function $v(t) = x'(t)$ must take the value 0 for some t between t_a and t_b .



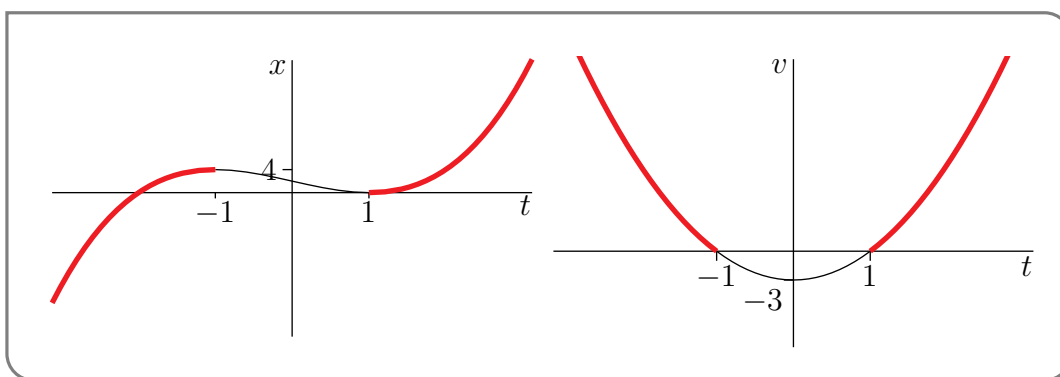
It is now easy to put together a mental image of your trajectory.

- For t large and negative (i.e. far in the past), $x(t)$ is large and negative and $v(t)$ is large and positive. For example³, when $t = -10^6$, $x(t) \approx t^3 = -10^{18}$ and $v(t) \approx 3t^2 = 3 \cdot 10^{12}$. So you are moving quickly to the right.
- For $t < -1$, $v(t) = x'(t) > 0$ so that $x(t)$ is increasing and you are moving to the right.
- At $t = -1$, $v(-1) = 0$ and you have come to a halt at position $x(-1) = (-1)^3 - 3(-1) + 2 = 4$.
- For $-1 < t < 1$, $v(t) = x'(t) < 0$ so that $x(t)$ is decreasing and you are moving to the left.
- At $t = +1$, $v(1) = 0$ and you have again come to a halt, but now at position $x(1) = 1^3 - 3 + 2 = 0$.
- For $t > 1$, $v(t) = x'(t) > 0$ so that $x(t)$ is increasing and you are again moving to the right.
- For t large and positive (i.e. in the far future), $x(t)$ is large and positive and $v(t)$ is large and positive. For example⁴, when $t = 10^6$, $x(t) \approx t^3 = 10^{18}$ and $v(t) \approx 3t^2 = 3 \cdot 10^{12}$. So you are moving quickly to the right.

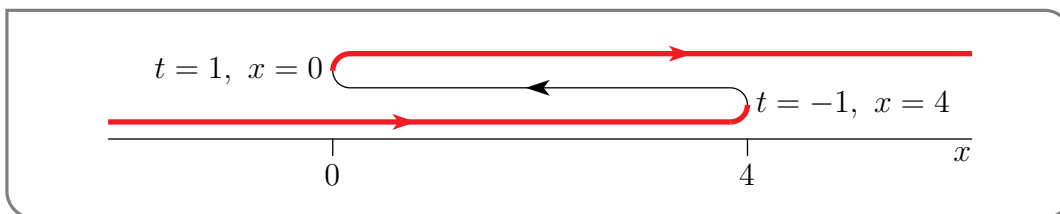
Here is a sketch of the graphs of $x(t)$ and $v(t)$. The heavy lines in the graphs indicate when you are moving to the right — that is where $v(t) = x'(t)$ is positive.

3 Notice here we are using the fact that when t is very large t^3 is much bigger than t^2 and t^1 . So we can approximate the value of the polynomial $x(t)$ by the largest term — in this case t^3 . We can do similarly with $v(t)$ — the largest term is $3t^2$.

4 We are making a similar rough approximation here.



And here is a schematic picture of the whole trajectory.



Example 3.1.1

Example 3.1.2

In this example we are going to figure out how far a body falling from rest will fall in a given time period.

- We should start by defining some variables and their units. Denote
 - time in seconds by t ,
 - mass in kilograms by m ,
 - distance fallen (in metres) at time t by $s(t)$, velocity (in m/sec) by $v(t) = s'(t)$ and acceleration (in m/sec²) by $a(t) = v'(t) = s''(t)$.

It makes sense to choose a coordinate system so that the body starts to fall at $t = 0$.

- We will use Newton's second law of motion

$$\text{the force applied to the body at time } t = m \cdot a(t).$$

together with the assumption that the only force acting on the body is gravity (in particular, no air resistance). Note that near the surface of the Earth,

$$\text{the force due to gravity acting on a body of mass } m = m \cdot g.$$

The constant g , called the acceleration of gravity⁵, is about 9.8m/sec².

5 It is also called the standard acceleration due to gravity or standard gravity. For those of you who prefer imperial units (or US customary units), it is about 32 ft/sec², 77165 cubits/minute², or 631353 furlongs/hour².

- Since the body is falling from rest, we know that its initial velocity is zero. That is

$$v(0) = 0.$$

Newton's second law then implies that

$$\begin{aligned} m \cdot a(t) &= \text{force due to gravity} \\ m \cdot v'(t) &= m \cdot g && \text{cancel the } m \\ v'(t) &= g \end{aligned}$$

- In order to find the velocity, we need to find a function of t whose derivative is constant. We are simply going to guess such a function and then we will verify that our guess has all of the desired properties. It's easy to guess a function whose derivative is the constant g . Certainly gt has the correct derivative. So does

$$v(t) = gt + c$$

for any constant c . One can then verify⁶ that $v'(t) = g$. Using the fact that $v(0) = 0$ we must then have $c = 0$ and so

$$v(t) = gt.$$

- Since velocity is the derivative of position, we know that

$$s'(t) = v(t) = g \cdot t.$$

To find $s(t)$ we are again going to guess and check. It's not hard to see that we can use

$$s(t) = \frac{g}{2}t^2 + c$$

where again c is some constant. Again we can verify that this works simply by differentiating⁷. Since we have defined $s(t)$ to be the distance fallen, it follows that $s(0) = 0$ which in turn tells us that $c = 0$. Hence

$$\begin{aligned} s(t) &= \frac{g}{2}t^2 && \text{but } g = 9.8, \text{ so} \\ &= 4.9t^2, \end{aligned}$$

which is exactly the $s(t)$ used way back in Section [1.2](#).

6 While it is clear that this satisfies the equation we want, it is less clear that it is the only function that works. To see this, assume that there are two functions $f(t)$ and $h(t)$ which both satisfy $v'(t) = g$. Then $f'(t) = h'(t) = g$ and so $f'(t) - h'(t) = 0$. Equivalently

$$\frac{d}{dt}(f(t) - h(t)) = 0.$$

The only function whose derivative is zero everywhere is the constant function (see Section [2.13](#) and Corollary [2.13.11](#)). Thus $f(t) - h(t) = \text{constant}$. So all the functions that satisfy $v'(t) = g$ must be of the form $gt + \text{constant}$.

7 To show that any solution of $s'(t) = gv$ must be of this form we can use the same reasoning we used to get $v(t) = gt + \text{constant}$.

Example 3.1.2

Let's now do a similar but more complicated example.

Example 3.1.3

A car's brakes can decelerate the car at 64000km/hr^2 . How fast can the car be driven if it must be able to stop within a distance of 50m ?

Solution. Before getting started, notice that there is a small “trick” in this problem — several quantities are stated but their units are different. The acceleration is stated in kilometres per hour², but the distance is stated in metres. Whenever we come across a “real world” problem⁸ we should be careful of the units used.

- We should first define some variables and their units. Denote
 - time (in hours) by t ,
 - the position of the car (in kilometres) at time t by $x(t)$, and
 - the velocity (in kilometres per hour) by $v(t)$.

We can also choose a coordinate system such that $x(0) = 0$ and the car starts braking at time $t = 0$.

- Now let us rewrite the information in the problem in terms of these variables.
 - We are told that, at maximum braking, the acceleration $v'(t) = x''(t)$ of the car is -64000 .
 - We need to determine the maximum initial velocity $v(0)$ so that the stopping distance is at most $50\text{m} = 0.05\text{km}$ (being careful with our units). Let us call the stopping distance x_{stop} which is really $x(t_{stop})$ where t_{stop} is the stopping time.
- In order to determine x_{stop} we first need to determine t_{stop} , which we will do by assuming maximum braking from a, yet to be determined, initial velocity of $v(0) = q$ m/sec.
- Assuming that the car undergoes a constant acceleration at this maximum braking power, we have

$$v'(t) = -64000$$

This equation is very similar to the ones we had to solve in Example 3.1.2 just above.

As we did there⁹, we are going to just guess $v(t)$. First, we just guess one function whose derivative is -64000 , namely $-64000t$. Next we observe that, since the derivative of a constant is zero, any function of the form

$$v(t) = -64000t + c$$

⁸ Well — “realer world” would perhaps be a betterer term.

⁹ Now is a good time to go back and have a read of that example.

with constant c , has the correct derivative. Finally, the requirement that the initial velocity $v(0) = q$ forces $c = q$, so

$$v(t) = q - 64000t$$

- From this we can easily determine the stopping time t_{stop} , when the initial velocity is q , since this is just when $v(t) = 0$:

$$0 = v(t_{stop}) = q - 64000 \cdot t_{stop} \quad \text{and so}$$

$$t_{stop} = \frac{q}{64000}.$$

- Armed with the stopping time, how do we get at the stopping distance? We need to find the formula satisfied by $x(t)$. Again (as per Example 3.1.2) we make use of the fact that

$$x'(t) = v(t) = q - 64000t.$$

So we need to guess a function $x(t)$ so that $x'(t) = q - 64000t$. It is not hard to see that

$$x(t) = qt - 32000t^2 + \text{constant}$$

works. Since we know that $x(0) = 0$, this constant is just zero and

$$x(t) = qt - 32000t^2.$$

- We are now ready to compute the stopping distance (in terms of the, still yet to be determined, initial velocity q):

$$\begin{aligned} x_{stop} &= x(t_{stop}) = qt_{stop} - 32000t_{stop}^2 \\ &= \frac{q^2}{64000} - \frac{32000q^2}{64000^2} \\ &= \frac{q^2}{64000} \left(1 - \frac{1}{2}\right) \\ &= \frac{q^2}{2 \times 64000} \end{aligned}$$

Notice that the stopping distance is a quadratic function of the initial velocity — if you go twice as fast, you need four times the distance to stop.

- But we are told that the stopping distance must be less than $50m = 0.05km$. This means that

$$\begin{aligned} x_{stop} &= \frac{q^2}{2 \times 64000} \leq \frac{5}{100} \\ q^2 &\leq \frac{2 \times 64000 \times 5}{100} = \frac{64000 \times 10}{100} = 6400 \end{aligned}$$

Thus we must have $q \leq 80$. Hence the initial velocity can be no greater than $80km/h$.

Example 3.1.3

3.2 ▲ Related Rates

Consider the following problem

A spherical balloon is being inflated at a rate of $13\text{cm}^3/\text{sec}$. How fast is the radius changing when the balloon has radius 15cm ?

There are several pieces of information in the statement:

- The balloon is spherical
- The volume is changing at a rate of $13\text{cm}^3/\text{sec}$ — so we need variables for volume (in cm^3) and time (in sec). Good choices are V and t .
- We are asked for the rate at which the radius is changing — so we need a variable for radius and units. A good choice is r , measured in cm — since volume is measured in cm^3 .

Since the balloon is a sphere we know¹⁰ that

$$V = \frac{4}{3}\pi r^3$$

Since both the volume and radius are changing with time, both V and r are implicitly functions of time; we could really write

$$V(t) = \frac{4}{3}\pi r(t)^3$$

We are told the rate at which the volume is changing and we need to find the rate at which the radius is changing. That is, from a knowledge of $\frac{dV}{dt}$, find the related rate¹¹ $\frac{dr}{dt}$.

In this case, we can just differentiate our equation by t to get

$$\frac{dV}{dt} = 4\pi r^2 \frac{dr}{dt}$$

This can then be rearranged to give

$$\frac{dr}{dt} = \frac{1}{4\pi r^2} \frac{dV}{dt}$$

Now we were told that $\frac{dV}{dt} = 13$, so

$$\frac{dr}{dt} = \frac{13}{4\pi r^2}$$

We were also told that the radius is 15cm , so at that moment in time

$$\frac{dr}{dt} = \frac{13}{\pi 4 \times 15^2}$$

This is a very typical example of a related rate problem. This section is really just a collection of problems, but all will follow a similar pattern.

10 If you don't know the formula for the volume of a sphere, now is a good time to revise by looking at Appendix [A.11](#)

11 Related rate problems are problems in which you are given the rate of change of one quantity and are to determine the rate of change of another, related, quantity.

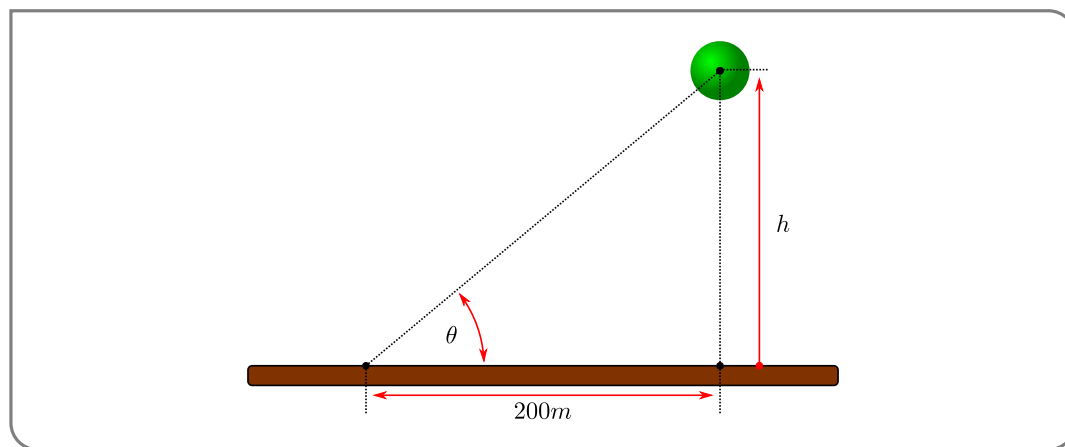
- The statement of the problem will tell you quantities that must be related (above it was volume, radius and, implicitly, time).
- Typically a little geometry (or some physics or...) will allow you to relate these quantities (above it was the formula that links the volume of a sphere to its radius).
- Implicit differentiation will then allow you to link the rate of change of one quantity to another.

Another balloon example

Example 3.2.1

Consider a helium balloon rising vertically from a fixed point 200m away from you. You are trying to work out how fast it is rising. Now — computing the velocity directly is difficult, but you can measure angles. You observe that when it is at an angle of $\pi/4$ its angle is changing by 0.05 radians per second.

- Start by drawing a picture with the relevant variables



- So denote the angle to be θ (in radians), the height of the balloon (in m) by h and time (in seconds) by t . Then trigonometry tells us

$$h = 200 \cdot \tan \theta$$

- Differentiating allows us to relate the rates of change

$$\frac{dh}{dt} = 200 \sec^2 \theta \cdot \frac{d\theta}{dt}$$

- We are told that when $\theta = \pi/4$ we observe $\frac{d\theta}{dt} = 0.05$, so

$$\begin{aligned} \frac{dh}{dt} &= 200 \cdot \sec^2(\pi/4) \cdot 0.05 \\ &= 200 \cdot 0.05 \cdot (\sqrt{2})^2 \\ &= 200 \cdot \frac{5}{100} \cdot 2 \\ &= 20m/s \end{aligned}$$

- So the balloon is rising at a rate of 20m/s.

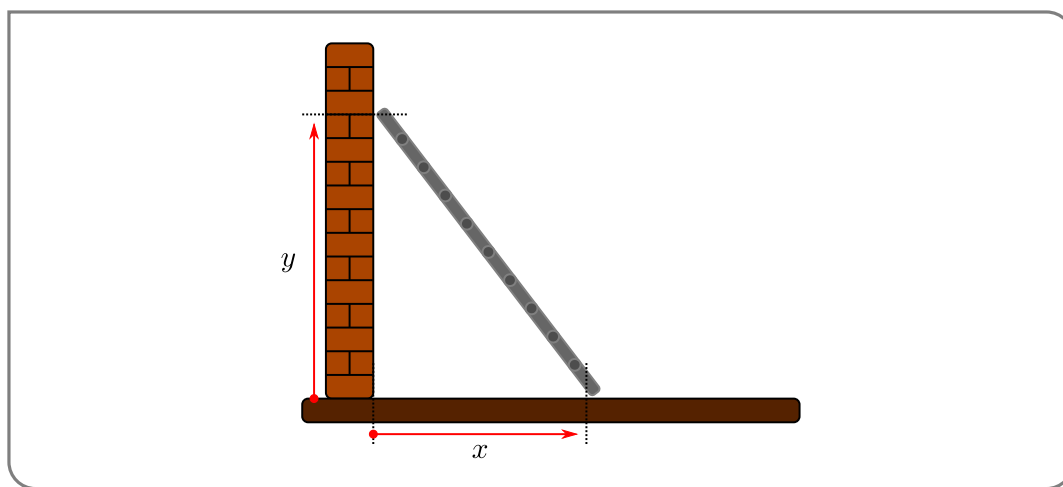
Example 3.2.1

The following problem is perhaps *the* classic related rate problem.

Example 3.2.2

A 5m ladder is leaning against a wall. The floor is quite slippery and the base of the ladder slides out from the wall at a rate of 1m/s. How fast is the top of the ladder sliding down the wall when the base of the ladder is 3m from the wall?

- A good first step is to draw a picture stating all relevant quantities. This will also help us define variables and units.



- So now define $x(t)$ to be the distance between the bottom of the ladder and the wall, at time t , and let $y(t)$ be the distance between the top of the ladder and the ground at time t . Measure time in seconds, but both distances in meters.
- We can relate the quantities using Pythagoras:

$$x^2 + y^2 = 5^2$$

- Differentiating with respect to time then gives

$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

- We know that $\frac{dx}{dt} = 1$ and $x = 3$, so

$$6 \cdot 1 + 2y \frac{dy}{dt} = 0$$

but we need to determine y before we can go further. Thankfully we know that $x^2 + y^2 = 25$ and $x = 3$, so $y^2 = 25 - 9 = 16$ and $y = 4$.

12 Since the ladder isn't buried in the ground, we can discard the solution $y = -4$.

- So finally putting everything together

$$6 \cdot 1 + 8 \frac{dy}{dt} = 0$$

$$\frac{dy}{dt} = -\frac{3}{4} m/s.$$

Thus the top of the ladder is sliding towards the floor at a rate of $3/4 m/s$.

Example 3.2.2

The next example is complicated by the rates of change being stated not just as “the rate of change per unit time” but instead being stated as “the percentage rate of change per unit time”. If a quantity f is changing with rate $\frac{df}{dt}$, then we can say that

f is changing at a rate of $100 \cdot \frac{\frac{df}{dt}}{f}$ percent.

Thus if, at time t , f has rate of change $r\%$, then

$$100 \frac{f'(t)}{f(t)} = r \implies f'(t) = \frac{r}{100} f(t)$$

so that if h is a very small time increment

$$\frac{f(t+h) - f(t)}{h} \approx \frac{r}{100} f(t) \implies f(t+h) \approx f(t) + \frac{rh}{100} f(t)$$

That is, over a very small time interval h , f increases by the fraction $\frac{rh}{100}$ of its value at time t .

So armed with this, let’s look at the problem.

Example 3.2.3

The quantities P , Q and R are functions of time and are related by the equation $R = PQ$. Assume that P is increasing instantaneously at the rate of 8% per year (meaning that $100 \frac{P'}{P} = 8$) and that Q is decreasing instantaneously at the rate of 2% per year (meaning that $100 \frac{Q'}{Q} = -2$). Determine the percentage rate of change for R .

Solution. This one is a little different — we are given the variables and the formula, so no picture drawing or defining required. Though we do need to define a time variable — let t denote time in years.

- Since $R(t) = P(t) \cdot Q(t)$ we can differentiate with respect to t to get

$$\frac{dR}{dt} = PQ' + QP'$$

- But we need the percentage change in R , namely

$$100 \frac{R'}{R} = 100 \frac{PQ' + QP'}{R}$$

but $R = PQ$, so rewrite it as

$$\begin{aligned} &= 100 \frac{PQ' + QP'}{PQ} \\ &= 100 \frac{PQ'}{PQ} + 100 \frac{QP'}{PQ} \\ &= 100 \frac{Q'}{Q} + 100 \frac{P'}{P} \end{aligned}$$

so we have stated the instantaneous percentage rate of change in R as the sum of the percentage rate of change in P and Q .

- We know the percentage rate of change of P and Q , so

$$100 \frac{R'}{R} = -2 + 8 = 6$$

That is, the instantaneous percentage rate of change of R is 6% per year.

Example 3.2.3

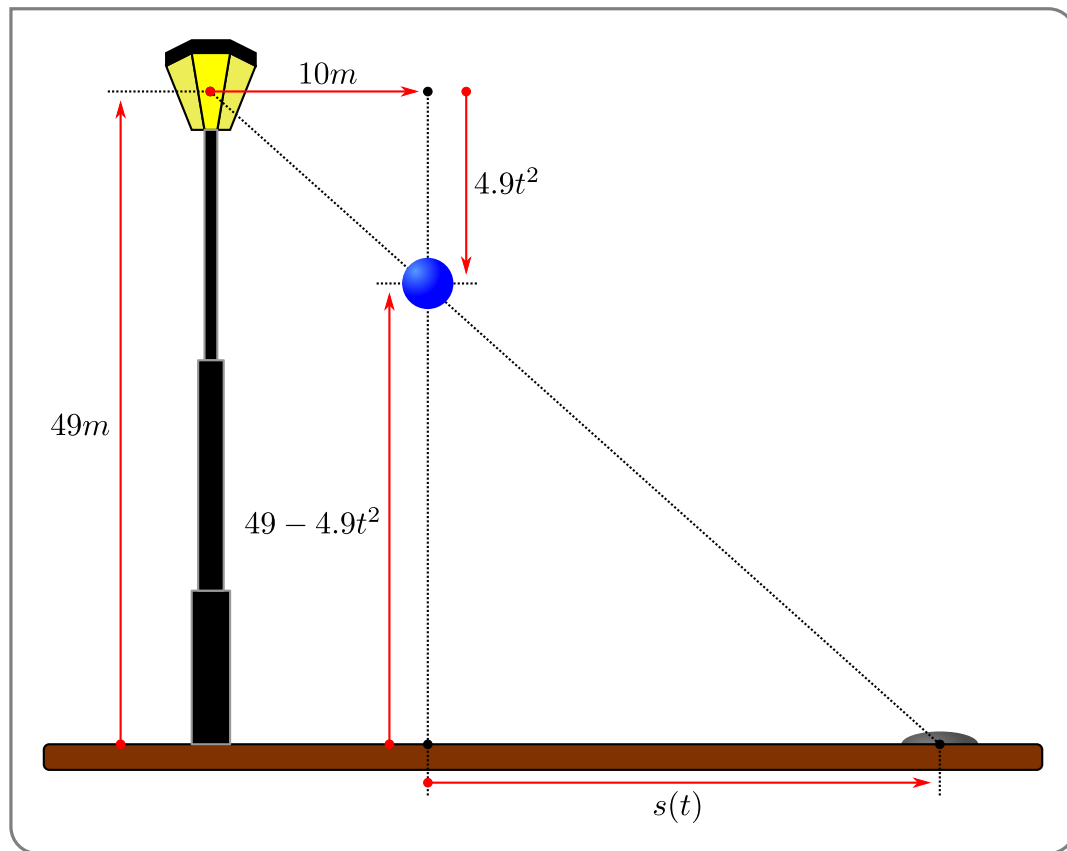
Yet another falling object example.

Example 3.2.4

A ball is dropped from a height of 49m above level ground. The height of the ball at time t is $h(t) = 49 - 4.9t^2$ m. A light, which is also 49m above the ground, is 10m to the left of the ball's original position. As the ball descends, the shadow of the ball caused by the light moves across the ground. How fast is the shadow moving one second after the ball is dropped?

Solution. There is quite a bit going on in this example, so read carefully.

- First a diagram; the one below is perhaps a bit over the top.



- Let's call $s(t)$ the distance from the shadow to the point on the ground directly underneath the ball.
- By similar triangles we see that

$$\frac{4.9t^2}{10} = \frac{49 - 4.9t^2}{s(t)}$$

We can then solve for $s(t)$ by just multiplying both sides by $\frac{10}{4.9t^2}s(t)$. This gives

$$s(t) = 10 \frac{49 - 4.9t^2}{4.9t^2} = \frac{100}{t^2} - 10$$

- Differentiating with respect to t will then give us the rates,

$$s'(t) = -2 \frac{100}{t^3}$$

- So, at $t = 1$, $s'(1) = -200\text{m/sec}$. That is, the shadow is moving to the left at 200m/sec.

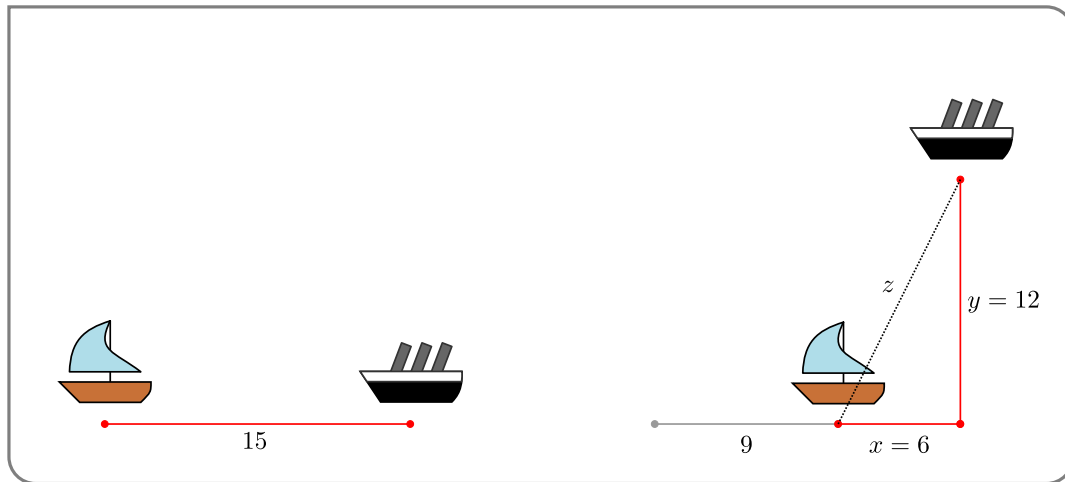
Example 3.2.4

A more nautical example.

Example 3.2.5

Two boats spot each other in the ocean at midday — boat A is 15km west of boat B . Boat A is travelling east at 3km/h and boat B is travelling north at 4km/h. How fast is the distance between the boats changing at 3pm?

- First we draw a picture.



- Let $x(t)$ be the distance at time t , in km, from boat A to the original position of boat B (i.e. to the position of boat B at noon). And let $y(t)$ be the distance at time t , in km, of boat B from its original position. And let $z(t)$ be the distance between the two boats at time t .
- Additionally we are told that $x' = -3$ and $y' = 4$ — notice that $x' < 0$ since that distance is getting smaller with time, while $y' > 0$ since that distance is increasing with time.
- Further at 3pm boat A has travelled 9km towards the original position of boat B , so $x = 15 - 9 = 6$, while boat B has travelled 12km away from its original position, so $y = 12$.
- The distances x, y and z form a right-angled triangle, and Pythagoras tells us that

$$z^2 = x^2 + y^2.$$

At 3pm we know $x = 6, y = 12$ so

$$z^2 = 36 + 144 = 180$$

$$z = \sqrt{180} = 6\sqrt{5}.$$

- Differentiating then gives

$$\begin{aligned} 2z \frac{dz}{dt} &= 2x \frac{dx}{dt} + 2y \frac{dy}{dt} \\ &= 12 \cdot (-3) + 24 \cdot (4) \\ &= 60. \end{aligned}$$

Dividing through by $2z = 12\sqrt{5}$ then gives

$$\frac{dz}{dt} = \frac{60}{12\sqrt{5}} = \frac{5}{\sqrt{5}} = \sqrt{5}$$

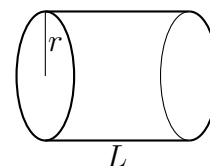
So the distance between the boats is increasing at $\sqrt{5} \text{ km/h}$.

Example 3.2.5

One last one before we move on to another topic.

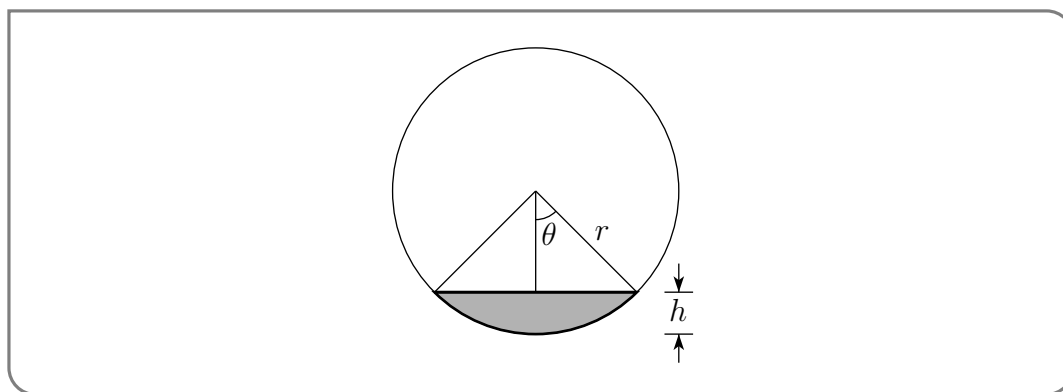
Example 3.2.6

Consider a cylindrical fuel tank of radius r and length L (in some appropriate units) that is lying on its side. Suppose that fuel is being pumped into the tank at a rate q . At what rate is the fuel level rising?



Solution. If the tank were vertical everything would be much easier. Unfortunately the tank is on its side, so we are going to have to work a bit harder to establish the relation between the depth and volume. Also notice that we have not been supplied with units for this problem — so we do not need to state the units of our variables.

- Again — draw a picture. Here is an end view of the tank; the shaded part of the circle is filled with fuel.



- Let us denote by $V(t)$ the volume of fuel in the tank at time t and by $h(t)$ the fuel level at time t .
- We have been told that $V'(t) = q$ and have been asked to determine $h'(t)$. While it is possible to do so by finding a formula relating $V(t)$ and $h(t)$, it turns out to be quite a bit easier to first find a formula relating V and the angle θ shown in the end view. We can then translate this back into a formula in terms of h using the relation

$$h(t) = r - r \cos \theta(t).$$

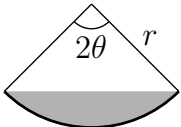
Once we know $\theta'(t)$, we can easily obtain $h'(t)$ by differentiating the above equation.

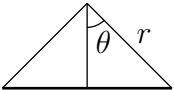
- The computation that follows below gets a little involved in places, so we will drop the “(t)” on the variables V, h and θ . The reader must never forget that these three quantities are really functions of time, while r and L are constants that do not depend on time.
- The volume of fuel is L times the cross-sectional area filled by the fuel. That is,

$$V = L \times \text{Area}(\text{☾})$$

While we do not have a canned formula for the area of a chord of a circle like this, it is easy to express the area of the chord in terms of two areas that we can compute.

$$V = L \times \text{Area}(\text{☾}) = L \times \left[\text{Area}\left(\text{☾} \text{ with angle } 2\theta \text{ and radius } r\right) - \text{Area}\left(\text{triangle with angle } \theta \text{ and radius } r\right) \right]$$

- The piece of pie  is the fraction $\frac{2\theta}{2\pi}$ of the full circle, so its area is $\frac{2\theta}{2\pi} \pi r^2 = \theta r^2$.

- The triangle  has height $r \cos \theta$ and base $2r \sin \theta$ and hence has area $\frac{1}{2}(r \cos \theta)(2r \sin \theta) = r^2 \sin \theta \cos \theta = \frac{r^2}{2} \sin(2\theta)$, where we have used a double-angle formula (see Appendix [A.14](#)).

Subbing these two areas into the above expression for V gives

$$V = L \times \left[\theta r^2 - \frac{r^2}{2} \sin 2\theta \right] = \frac{Lr^2}{2} [2\theta - \sin 2\theta]$$

Oof!

- Now we can differentiate to find the rate of change. Recalling that $V = V(t)$ and $\theta = \theta(t)$, while r and L are constants,

$$\begin{aligned} V' &= \frac{Lr^2}{2} [2\theta' - 2 \cos 2\theta \cdot \theta'] \\ &= Lr^2 \cdot \theta' \cdot [1 - \cos 2\theta] \end{aligned}$$

Solving this for θ' and using $V' = q$ gives

$$\theta' = \frac{q}{Lr^2(1 - \cos 2\theta)}$$

This is the rate at which θ is changing, but we need the rate at which h is changing. We get this from

$$\begin{aligned} h &= r - r \cos \theta & \text{differentiating this gives} \\ h' &= r \sin \theta \cdot \theta' \end{aligned}$$

APPLICATIONS OF DERIVATIVES

3.1▲ Velocity and acceleration

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Suppose you throw a ball straight up in the air, and its height from $t = 0$ to $t = 4$ is given by $h(t) = -4.9t^2 + 19.6t$. True or false: at time $t = 2$, the acceleration of the ball is 0.

Q[2]: Suppose an object is moving with a constant acceleration. It takes ten seconds to accelerate from $1 \frac{\text{m}}{\text{s}}$ to $2 \frac{\text{m}}{\text{s}}$. How long does it take to accelerate from $2 \frac{\text{m}}{\text{s}}$ to $3 \frac{\text{m}}{\text{s}}$? How long does it take to accelerate from $3 \frac{\text{m}}{\text{s}}$ to $13 \frac{\text{m}}{\text{s}}$?

Q[3]: Let $s(t)$ be the position of a particle at time t . True or false: if $s''(a) > 0$ for some a , then the particle's speed is increasing when $t = a$.

Q[4]: Let $s(t)$ be the position of a particle at time t . True or false: if $s'(a) > 0$ and $s''(a) > 0$ for some a , then the particle's speed is increasing when $t = a$.

►► Stage 2

For this section, we will ask you a number of questions that have to do with objects falling on Earth. Unless otherwise stated, you should assume that an object falling through the air has an acceleration due to gravity of 9.8 meters per second per second.

Q[5]: A flower pot rolls out of a window 10m above the ground. How fast is it falling just as it smacks into the ground?

Q[6]: You want to know how deep a well is, so you drop a stone down and count the seconds until you hear it hit bottom.

- (a) If the stone took x seconds to hit bottom, how deep is the well?
- (b) Suppose you think you dropped the stone down the well, but actually you *tossed* it down, so instead of an initial velocity of 0 metres per second, you accidentally imparted an initial speed of 1 metres per second. What is the actual depth of the well, if the stone fell for x seconds?

Q[7]: You toss a key to your friend, standing two metres away. The keys initially move towards your friend at 2 metres per second, but slow at a rate of 0.25 metres per second per second. How much time does your friend have to react to catch the keys? That is—how long are the keys flying before they reach your friend?

Q[8]: A car is driving at 100 kph, and it brakes with a deceleration of $50\,000 \frac{\text{km}}{\text{hr}^2}$. How long does the car take to come to a complete stop?

Q[9]: You are driving at 120 kph, and need to stop in 100 metres. How much deceleration do your brakes need to provide? You may assume the brakes cause a constant deceleration.

Q[10]: You are driving at 100 kph, and apply the brakes steadily, so that your car decelerates at a constant rate and comes to a stop in exactly 7 seconds. What was your speed one second before you stopped?

Q[11]: About 8.5 minutes after liftoff, the US space shuttle has reached orbital velocity, 17 500 miles per hour. Assuming its acceleration was constant, how far did it travel in those 8.5 minutes?

(Source: https://www.nasa.gov/mission_pages/shuttle/shuttlemissions/sts121/launch/qa-leinbach.html)

Q[12]: A pitching machine has a dial to adjust the speed of the pitch. You rotate it so that it pitches the ball straight up in the air. How fast should the ball exit the machine, in order to stay in the air exactly 10 seconds?

You may assume that the ball exits from ground level, and is acted on only by gravity, which causes a constant deceleration of 9.8 metres per second.

Q[13]: A peregrine falcon can dive at a speed of 325 kph. If you were to drop a stone, how high up would you have to be so that the stone reached the same speed in its fall?

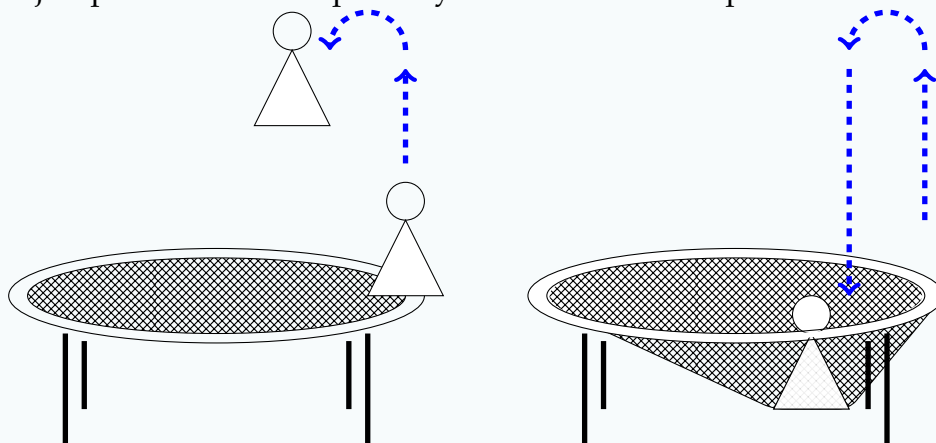
Q[14]: You shoot a cannon ball into the air with initial velocity v_0 , and then gravity brings it back down (neglecting all other forces). If the cannon ball made it to a height of 100m, what was v_0 ?

Q[15]: Suppose you are driving at 120 kph, and you start to brake at a deceleration of 50000

kph per hour. For three seconds you steadily increase your deceleration to 60000 kph per hour. (That is, for three seconds, the rate of change of your deceleration is constant.) How fast are you driving at the end of those three seconds?

►► Stage 3

Q[16]: You jump up from the side of a trampoline with an initial upward velocity of 1 metre per second. While you are in the air, your deceleration is a constant 9.8 metres per second per second due to gravity. Once you hit the trampoline, as you fall your speed decreases by 4.9 metres per second per second. How many seconds pass between the peak of your jump and the lowest part of your fall on the trampoline?



Q[17]: Suppose an object is moving so that its velocity doubles every second. Give an expression for the acceleration of the object.

3.2▲ Related Rates

Exercises

Jump to [HINTS](#), [ANSWERS](#), [SOLUTIONS](#) or [TABLE OF CONTENTS](#).

►► Stage 1

Q[1]: Suppose the quantities P and Q are related by the formula $P = Q^3$. P and Q are changing with respect to time, t . Given this information, which of the following are problems you could solve?

- Given $\frac{dP}{dt}(0)$, find $\frac{dQ}{dt}(0)$. (Remember: the notation $\frac{dP}{dt}(0)$ means the derivative of P with respect to t at the time $t = 0$.)
- Given $\frac{dP}{dt}(0)$ and the value of Q when $t = 0$, find $\frac{dQ}{dt}(0)$.
- Given $\frac{dQ}{dt}(0)$, find $\frac{dP}{dt}(0)$.
- Given $\frac{dQ}{dt}(0)$ and the value of P when $t = 0$, find $\frac{dP}{dt}(0)$.

►► Stage 2

For problems 2 through 4, the relationship between several variables is explicitly given. Use this information to relate their rates of change.

Q[2](*) : A point is moving on the unit circle $\{(x, y) : x^2 + y^2 = 1\}$ in the xy -plane. At $(2/\sqrt{5}, 1/\sqrt{5})$, its y -coordinate is increasing at rate 3. What is the rate of change of its x -coordinate?

Q[3](*) : The quantities P , Q and R are functions of time and are related by the equation $R = PQ$. Assume that P is increasing instantaneously at the rate of 8% per year and that Q is decreasing instantaneously at the rate of 2% per year. That is, $\frac{P'}{P} = 0.08$ and $\frac{Q'}{Q} = -0.02$. Determine the percentage rate of change for R .

Q[4](*) : Three quantities, F , P and Q all depend upon time t and are related by the equation

$$F = \frac{P}{Q}$$

- Assume that at a particular moment in time $P = 25$ and P is increasing at the instantaneous rate of 5 units/min. At the same moment, $Q = 5$ and Q is increasing at the instantaneous rate of 1 unit/min. What is the instantaneous rate of change in F at this moment?
- Assume that at another moment in time P is increasing at the instantaneous rate of 10% and Q is decreasing at the instantaneous rate 5%. What can you conclude about the rate of change of F at this moment?

For Questions 5 through 9, look for a way to use the Pythagorean Theorem.

Q[5](*) : Two particles move in the Cartesian plane. Particle A travels on the x -axis starting at $(10, 0)$ and moving towards the origin with a speed of 2 units per second. Particle B

travels on the y -axis starting at $(0, 12)$ and moving towards the origin with a speed of 3 units per second. What is the rate of change of the distance between the two particles when particle A reaches the point $(4, 0)$?

Q[6](*) : Two particles A and B are placed on the Cartesian plane at $(0, 0)$ and $(3, 0)$ respectively. At time 0, both start to move in the $+y$ direction. Particle A moves at 3 units per second, while B moves at 2 units per second. How fast is the distance between the particles changing when particle A is at a distance of 5 units from B .

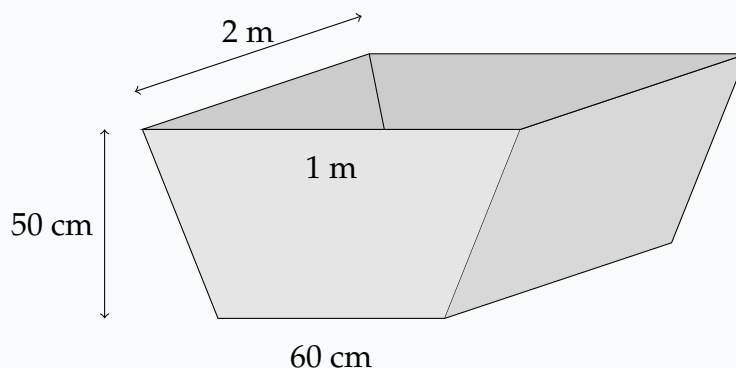
Q[7](*) : Ship A is 400 miles directly south of Hawaii and is sailing south at 20 miles/hour. Ship B is 300 miles directly east of Hawaii and is sailing west at 15 miles/hour. At what rate is the distance between the ships changing?

Q[8](*) : Two tall sticks are vertically planted into the ground, separated by a distance of 30 cm. We simultaneously put two snails at the base of each stick. The two snails then begin to climb their respective sticks. The first snail is moving with a speed of 25 cm per minute, while the second snail is moving with a speed of 15 cm per minute. What is the rate of change of the distance between the two snails when the first snail reaches 100 cm above the ground?

Q[9](*) : A 20m long extension ladder leaning against a wall starts collapsing in on itself at a rate of 2m/s, while the foot of the ladder remains a constant 5m from the wall. How fast is the ladder moving down the wall after 3.5 seconds?

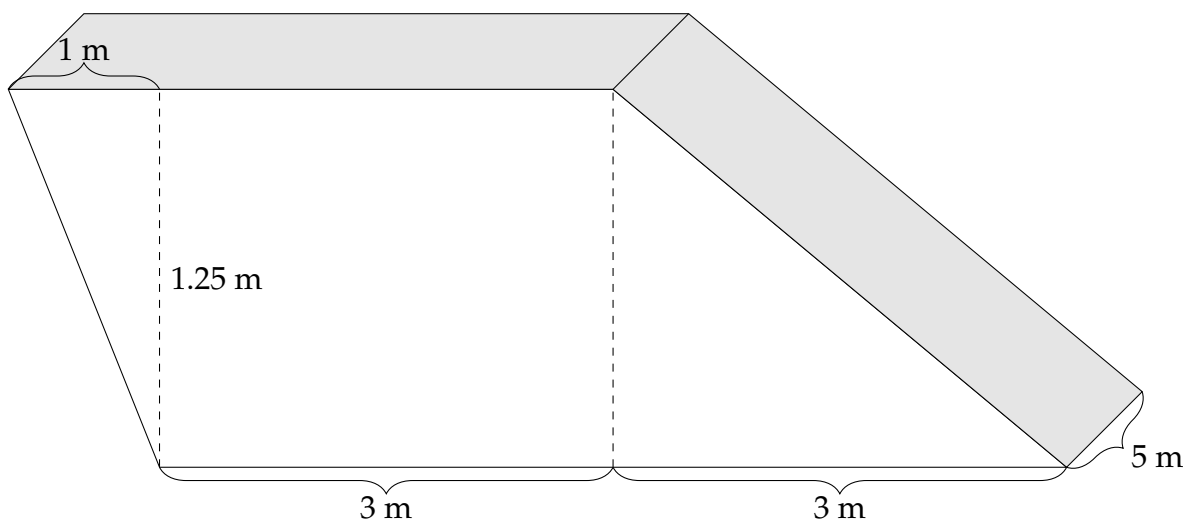
For Questions 10 through 14, look for tricks from trigonometry.

Q[10]: A watering trough has a cross section shaped like an isosceles trapezoid. The trough is 2 metres long, 50 cm high, 1 metre wide at the top, and 60 cm wide at the bottom.



A pig is drinking water from the trough at a rate of 3 litres per minute. When the height of the water is 25 cm, how fast is the height decreasing?

Q[11]: A tank is 5 metres long, and has a trapezoidal cross section with the dimensions shown below.



A hose is filling the tank up at a rate of one litre per second. How fast is the height of the water increasing when the water is 10 centimetres deep?

Q[12]: A rocket is blasting off, 2 kilometres away from you. You and the rocket start at the same height. The height of the rocket in kilometres, t hours after liftoff, is given by

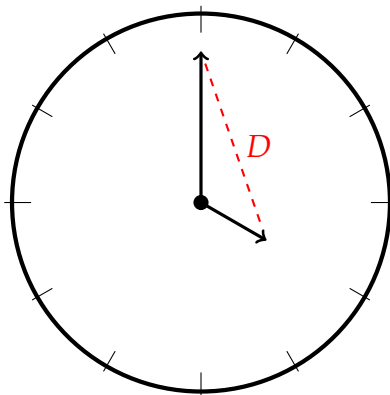
$$h(t) = 61750t^2$$

How fast (in radians per second) is your line of sight rotating to keep looking at the rocket, one minute after liftoff?

Q[13](*): A high speed train is traveling at 2 km/min along a straight track. The train is moving away from a movie camera which is located 0.5 km from the track.

- How fast is the distance between the train and the camera increasing when they are 1.3 km apart?
- Assuming that the camera is always pointed at the train, how fast (in radians per min) is the camera rotating when the train and the camera are 1.3 km apart?

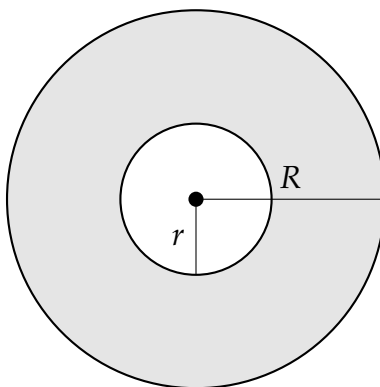
Q[14]: A clock has a minute hand that is 10 cm long, and an hour hand that is 5 cm long. Let D be the distance between the tips of the two hands. How fast is D decreasing at 4:00?



For Questions [15](#) through [20](#), you'll need to know formulas for volume or area.

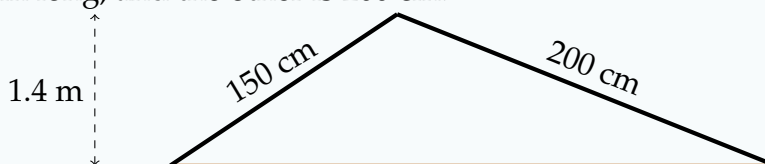
Q[15](*) Find the rate of change of the area of the annulus $\{(x, y) : r^2 \leq x^2 + y^2 \leq R^2\}$. (i.e. the points inside the circle of radius R but outside the circle of radius r) if

$$R = 3 \text{ cm}, r = 1 \text{ cm}, \frac{dR}{dt} = 2 \frac{\text{cm}}{\text{s}}, \text{ and } \frac{dr}{dt} = 7 \frac{\text{cm}}{\text{s}}.$$



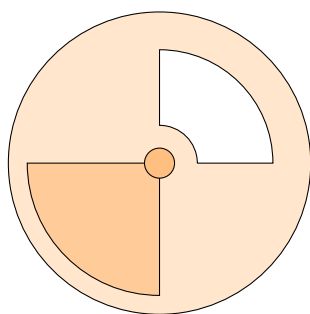
Q[16]: Two spheres are centred at the same point. The radius R of the bigger sphere at time t is given by $R(t) = 10 + 2t$, while the radius r of the smaller sphere is given by $r(t) = 6t$, $t \geq 0$. How fast is the volume between the spheres (inside the big sphere and outside the small sphere) changing when the bigger sphere has a radius twice as large as the smaller?

Q[17]: You attach two sticks together at their ends, and stick the other ends in the mud. One stick is 150 cm long, and the other is 200 cm.

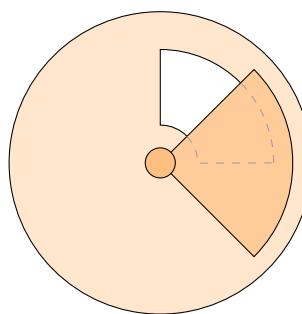


The structure starts out being 1.4 metres high at its peak, but the sticks slide, and the height decreases at a constant rate of three centimetres per minute. How quickly is the area of the triangle (formed by the two sticks and the level ground) changing when the height of the structure is 120 cm?

Q[18]: The circular lid of a salt shaker has radius 8. There is a cut-out to allow the salt to pour out of the lid, and a door that rotates around to cover the cut-out. The door is a quarter-circle of radius 7 cm. The cut-out has the shape of a quarter-annulus with outer radius 6 cm and inner radius 1 cm. If the uncovered area of the cut-out is $A \text{ cm}^2$, then the salt flows out at $\frac{1}{5}A \text{ cm}^3$ per second.

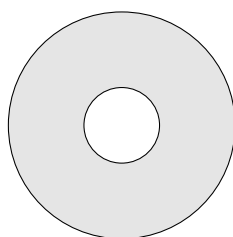


salt shaker lid
cut-out uncovered

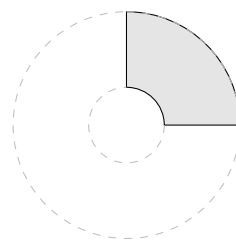


salt shaker lid
cut-out partially covered

Recall: an annulus is the set of points inside one circle and outside another, like a flat doughnut (see Question [15](#)).



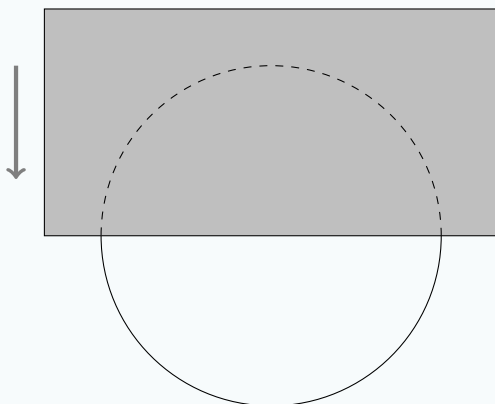
annulus



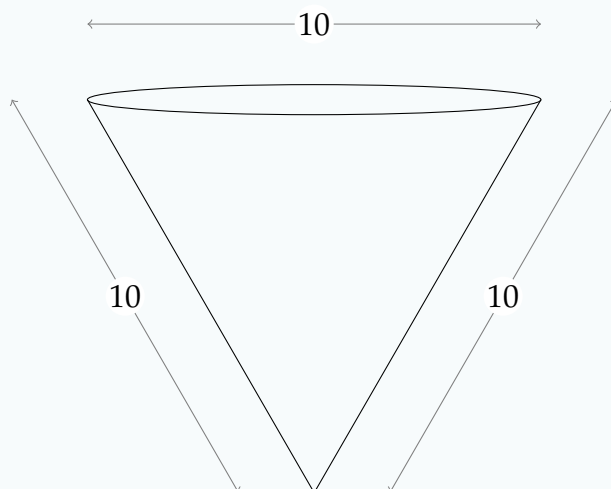
quarter annulus

While pouring out salt, you spin the door around the lid at a constant rate of $\frac{\pi}{6}$ radians per second, covering more and more of the cut-out. When exactly half of the cut-out is covered, how fast is the flow of salt changing?

Q[19]: A cylindrical sewer pipe with radius 1 metre has a vertical rectangular door that slides in front of it to block the flow of water, as shown below. If the uncovered area of the pipe is A m², then the flow of water through the pipe is $\frac{1}{5}A$ cubic metres per second. The door slides over the pipe, moving vertically at a rate of 1 centimetre per second. How fast is the flow of water changing when the door covers the top 25 centimetres of the pipe?



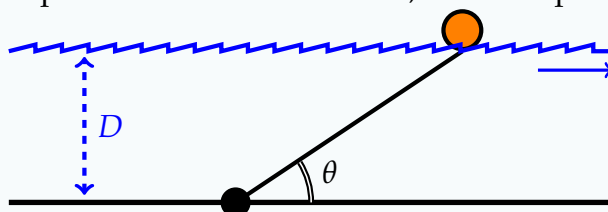
Q[20]: A martini glass is shaped like a cone, with top diameter 10 cm and side length 10 cm.



When the liquid in the glass is 7 cm high, it is evaporating at a rate of 5 mL per minute. How fast is the height of the liquid decreasing?

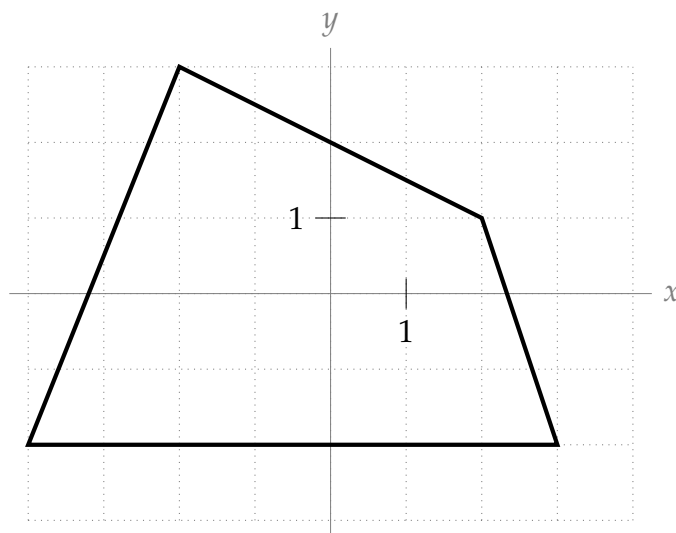
► Stage 3

Q[21]: A floating buoy is anchored to the bottom of a river. As the river flows, the buoy is pulled in the direction of flow until its 2-metre rope is taut. A sensor at the anchor reads the angle θ between the rope and the riverbed, as shown in the diagram below. This data is used to measure the depth D of water in the river, which depends on time.



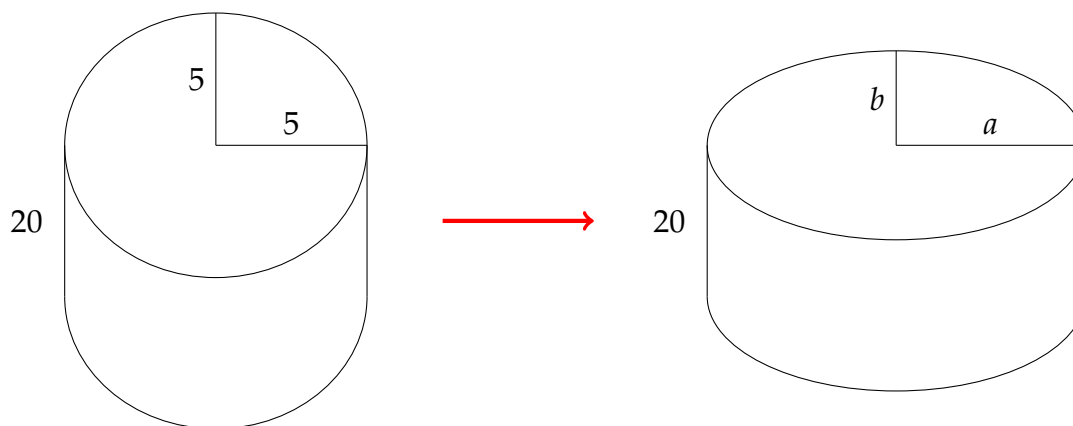
- If $\theta = \frac{\pi}{4}$ and $\frac{d\theta}{dt} = 0.25 \frac{\text{rad}}{\text{hr}}$, how fast is the depth D of the water changing?
- A measurement shows $\frac{d\theta}{dt} = 0$, but $\frac{dD}{dt} \neq 0$. Under what circumstances does this occur?
- A measurement shows $\frac{d\theta}{dt} > 0$, but $\frac{dD}{dt} < 0$. Under what circumstances does this occur?

Q[22]: A point is moving in the xy -plane along the quadrilateral shown below.



- (a) When the point is at $(0, -2)$, it is moving to the right. An observer stationed at the origin must turn at a rate of one radian per second to keep looking directly at the point. How fast is the point moving?
- (b) When the point is at $(0, 2)$, its x -coordinate is increasing at a rate of one unit per second. How fast is its y -coordinate changing? How fast is the point moving?

Q[23]: You have a cylindrical water bottle 20 cm high, filled with water. Its cross section is a circle of radius 5. You slowly smoosh the sides, so the cross section becomes an ellipse with major axis (widest part) $2a$ and minor axis (skinniest part) $2b$.



After t seconds of smooshing the bottle, $a = 5 + t$ cm. The perimeter of the cross section is unchanged as the bottle deforms. The perimeter of an ellipse is actually quite difficult to calculate, but we will use an approximation derived by Ramanujan and assume that the perimeter p of our ellipse is

$$p \approx \pi \left[3(a + b) - \sqrt{(a + 3b)(3a + b)} \right].$$

The area of an ellipse is πab .

- (a) Give an equation that relates a and b (and no other variables).

and so f is differentiable at $x = 0$ and $f'(0) = 0$.

(b) From (a), $f'(0) = 0$ and

$$f(x) = \begin{cases} -x^2 & x < 0 \\ x^2 & x \geq 0. \end{cases}$$

So,

$$f'(x) = \begin{cases} -2x & x < 0 \\ 2x & x \geq 0. \end{cases}$$

Then, we know the second derivative of f everywhere except at $x = 0$:

$$f''(x) = \begin{cases} -2 & x < 0 \\ ?? & x = 0 \\ 2 & x > 0. \end{cases}$$

So, whenever $x \neq 0$, $f''(x)$ exists. To investigate the differentiability of $f'(x)$ when $x = 0$, again we turn to the definition of a derivative. If

$$\lim_{h \rightarrow 0} \frac{f'(0+h) - f'(0)}{h}$$

exists, then $f''(0)$ exists.

$$\lim_{h \rightarrow 0} \frac{f'(0+h) - f'(0)}{h} = \lim_{h \rightarrow 0} \frac{f'(h) - 0}{h} = \lim_{h \rightarrow 0} \frac{f'(h)}{h}$$

Since $f(h)$ behaves differently when h is greater than or less than zero, we look at the one-sided limits.

$$\begin{aligned} \lim_{h \rightarrow 0^+} \frac{f'(h)}{h} &= \lim_{h \rightarrow 0^+} \frac{2h}{h} = 2 \\ \lim_{h \rightarrow 0^-} \frac{f'(h)}{h} &= \lim_{h \rightarrow 0^-} \frac{-2h}{h} = -2 \end{aligned}$$

Since the one-sided limits do not agree,

$$\lim_{h \rightarrow 0} \frac{f'(0+h) - f'(0)}{h} = DNE$$

So, $f''(0)$ does not exist. Now we have a complete picture of $f''(x)$:

$$f''(x) = \begin{cases} -2 & x < 0 \\ DNE & x = 0 \\ 2 & x > 0. \end{cases}$$

Solutions to Exercises 3.1 — Jump to [TABLE OF CONTENTS](#)

S-1: False. The acceleration of the ball is given by $h''(t) = -9.8$. This is constant throughout its trajectory (and is due to gravity).

Remark: the *velocity* of the ball at $t = 2$ is zero, since $h'(2) = -9.8(2) + 19.6 = 0$, but the velocity is only zero for an instant. Since the velocity is changing, the acceleration is nonzero.

S-2: The acceleration is constant, which means the rate of change of the velocity is constant. So, since it took 10 seconds for the velocity to increase by 1 metre per second (from $1 \frac{\text{m}}{\text{s}}$ to $2 \frac{\text{m}}{\text{s}}$), then it *always* takes 10 seconds for the velocity to increase by 1 metre per second.

So, it takes 10 seconds to accelerate from $2 \frac{\text{m}}{\text{s}}$ to $3 \frac{\text{m}}{\text{s}}$. To accelerate from $3 \frac{\text{m}}{\text{s}}$ to $13 \frac{\text{m}}{\text{s}}$ (that is, to change its velocity by 10 metres per second), it takes $10 \times 10 = 100$ seconds.

S-3: Let $v(a) = s'(a)$ be the velocity of the particle. If $s''(a) > 0$, then $v'(a) > 0$ — so the *velocity* of the particle is increasing. However, that does not mean that its *speed* (the absolute value of velocity) is increasing as well. For example, if a velocity is increasing from -4 kph to -3 kph, the speed is decreasing from 4 kph to 3 kph. So, the statement is false in general.

Contrast this to Question **4**.

S-4: Since $s'(a) > 0$, $|s'(a)| = s'(a)$: that is, the speed and velocity of the particle are the same. (This means the particle is moving in the positive direction.) If $s''(a) > 0$, then the velocity (and hence speed) of the particle is increasing. So, the statement is true.

S-5: From Example 3.1.2, we know that an object falling from rest on the Earth is subject to the acceleration due to gravity, $9.8 \frac{\text{m}}{\text{s}^2}$. So, if $h(t)$ is the height of the flower pot t seconds after it rolls out the window, then $h''(t) = -9.8$. (We make the acceleration negative, since the measure “height” has “up” as the positive direction, while gravity pulls the pot in the negative direction, “down.”)

Then $h'(t)$ is a function whose derivative is the constant -9.8 and with $h'(0) = 0$ (since the object fell, instead of being thrown up or down), so $h'(t) = -9.8t$.

What we want to know is $h'(t)$ at the time t when the pot hits the ground. We don't know yet exactly what time that happens, so we go a little farther and find an expression for $h(t)$. The function $h(t)$ has derivative $-9.8t$ and $h(0) = 10$, so (again following the ideas in Example 3.1.2)

$$h(t) = -\frac{9.8}{2}t^2 + 10$$

Now, we can find the time when the pot hits the ground: it is the time when $h(t) = 0$

(and $t > 0$).

$$\begin{aligned}0 &= \frac{-9.8}{2}t^2 + 10 \\ \frac{9.8}{2}t^2 &= 10 \\ t^2 &= \frac{20}{9.8} \\ t &= +\sqrt{\frac{20}{9.8}} \approx 1.4 \text{ sec}\end{aligned}$$

The velocity of the pot at this time is

$$h' \left(\sqrt{\frac{20}{9.8}} \right) = -9.8 \left(\sqrt{\frac{20}{9.8}} \right) = -\sqrt{20 \cdot 9.8} = -14 \frac{\text{m}}{\text{s}}$$

So, the pot is falling at 14 metres per second, just as it hits the ground.

S-6: (a)

- Let $s(t)$ be the distance the stone has fallen t seconds after dropping it. Since the acceleration due to gravity is $9.8 \frac{\text{m}}{\text{s}^2}$, $s''(t) = 9.8$. (We don't make this negative, because $s(t)$ measures how far the stone has fallen, which means the positive direction in our coordinate system is "down," which is exactly the way gravity is pulling.)
- Then $s'(t)$ has a constant derivative of 9.8, so $s'(t) = 9.8t + c$ for some constant c . Notice $s'(0) = c$, so c is the velocity of the stone at the very instant you dropped it, which is zero. Therefore, $s'(t) = 9.8t$.
- So, $s(t)$ is a function with derivative $9.8t$. It's not too hard to figure out by guessing and checking that $s(t) = \frac{9.8}{2}t^2 + d$ for some constant d . Notice $s(0) = d$, so d is the distance the rock has travelled at the instant you dropped it, which is zero. So, $s(t) = \frac{9.8}{2}t^2 = 4.9t^2$.

Remark: this is exactly the formula found in Example 3.1.2. You may, in general, use that formula without proof, but you need to know where it comes from and be able to apply it in other circumstances where it might be slightly different—like part (b) below.

- The rock falls for x seconds, so the distance fallen is

$$4.9x^2$$

Remark: this is a decent (if imperfect) way to figure out how deep a well is, or how tall a cliff is, when you're out and about. Drop a rock, square the time, multiply by 5.

(b) We'll go through a similar process as before.

Again, let $s(t)$ be the distance the rock has fallen t seconds after it is let go. Then $s''(t) = 9.8$, so $s'(t) = 9.8t + c$. In this case, since the initial speed of the rock is 1 metre per second, $1 = s'(0) = c$, so $s'(t) = 9.8t + 1$.

Then, $s(t)$ is a function whose derivative is $9.8t + 1$, so $s(t) = \frac{9.8}{2}t^2 + t + d$ for some constant d . Since $0 = s(0) = d$, we see $s(t) = 4.9t^2 + t$.

So, if the rock falls for x seconds, the distance fallen is

$$4.9x^2 + x$$

Remark: This means there is an error of x metres in your estimation of the depth of the well.

S-7: Let $s(t)$ be the distance your keys have travelled since they left your hand. The rate at which they are travelling, $s'(t)$, is decreasing by 0.25 metres per second. That is, $s''(t) = -0.25$. Therefore, $s'(t) = -0.25t + c$ for some constant c . Since $c = s'(0) = 2$, we see

$$s'(t) = 2 - 0.25t$$

Then $s(t)$ has $2 - 0.25t$ as its derivative, so $s(t) = 2t - \frac{1}{8}t^2 + d$ for some constant d . At time $t = 0$, the keys have not yet gone anywhere, so $0 = s(0) = d$. Therefore,

$$s(t) = 2t - \frac{1}{8}t^2$$

The keys reach your friend when $s(t) = 2$ and $t > 0$. That is:

$$\begin{aligned} 2 &= 2t - \frac{1}{8}t^2 \\ 0 &= \frac{1}{8}t^2 - 2t + 2 \\ t &= 8 \pm 4\sqrt{3} \end{aligned}$$

We need to figure out which of these values of t is really the time when the keys reach your friend. The keys travel this way from $t = 0$ to the time they reach your friend. (Then $s(t)$ no longer describes their motion.) So, we need to find the first value of t that is positive with $s(t) = 2$. Since $8 - 4\sqrt{3} > 0$, this is the first time $s(t) = 2$ and $t > 0$. So, the keys take

$$8 - 4\sqrt{3} \approx 1 \text{ second}$$

to reach your friend.

S-8: We proceed with the technique of Example 3.1.3 in mind.

Let $v(t)$ be the velocity (in kph) of the car at time t , where t is measured in hours and $t = 0$ is the instant the brakes are applied. Then $v(0) = 100$ and $v'(t) = -50\,000$. Since $v'(t)$ is constant, $v(t)$ is a line with slope $-50\,000$ and intercept $(0, 100)$, so

$$v(t) = 100 - 50\,000t$$

The car comes to a complete stop when $v(t) = 0$, which occurs at $t = \frac{100}{50\,000} = \frac{1}{500}$ hours. This is a confusing measure, so we convert it to seconds:

$$\left(\frac{1}{500} \text{ hrs}\right) \left(\frac{3600 \text{ sec}}{1 \text{ hr}}\right) = 7.2 \text{ sec}$$

S-9: Suppose the deceleration provided by the brakes is $d \frac{\text{km}}{\text{hr}^2}$. Then if $v(t)$ is the velocity of the car, $v(t) = 120 - dt$ (at $t = 0$, the velocity is 120, and it decreases by d kph per hour). The car stops when $0 = v(t)$, so $t = \frac{120}{d}$ hours.

Let $s(t)$ be the distance the car has travelled t hours after applying the brakes. Then $s'(t) = v(t)$, so $s(t) = 120t - \frac{d}{2}t^2 + c$ for some constant c . Since $0 = s(0) = c$,

$$s(t) = 120t - \frac{d}{2}t^2$$

The car needs to stop in 100 metres, which is $\frac{1}{10}$ kilometres. We already found that the stopping time is $t = \frac{120}{d}$. So:

$$\begin{aligned} \frac{1}{10} &= s\left(\frac{120}{d}\right) \\ \frac{1}{10} &= 120\left(\frac{120}{d}\right) - \frac{d}{2}\left(\frac{120}{d}\right)^2 \end{aligned}$$

Multiplying both sides by d :

$$\begin{aligned} \frac{d}{10} &= 120^2 - \frac{120^2}{2} \\ d &= 5 \cdot 120^2 \end{aligned}$$

So, the brakes need to apply 72 000 kph per hour of deceleration.

S-10: Since your deceleration is constant, your speed decreases smoothly from 100 kph to 0 kph. So, one second before your stop, you only have $\frac{1}{7}$ of our speed left: you're going $\frac{100}{7}$ kph.

A less direct way to solve this problem is to note that $v(t) = 100 - dt$ is the velocity of car t hours after braking, if d is its deceleration. Since it stops in 7 seconds (or $\frac{7}{3600}$ hours), $0 = v\left(\frac{7}{3600}\right) = 100 - \frac{7}{3600}d$, so $d = \frac{360000}{7}$. Then

$$v\left(\frac{6}{3600}\right) = 100 - \left(\frac{360000}{7}\right)\left(\frac{6}{3600}\right) = 100 - \frac{6}{7} \cdot 100 = \frac{100}{7} \text{ kph}$$

S-11: If the acceleration was constant, then it was

$$\frac{17500 \text{ mph}}{\frac{8.5}{60} \text{ hr}} \approx 123500 \frac{\text{miles}}{\text{hr}^2}$$

So, the velocity t hours from liftoff is

$$v(t) = 123500t$$

Therefore, the position of the shuttle t hours from liftoff (taking $s(0) = 0$ to be its initial position) is

$$s(t) = \frac{123500}{2}t^2 = 61750t^2$$

So, after $\frac{8.5}{60}$ hours, the shuttle has travelled

$$s\left(\frac{8.5}{60}\right) = (61750)\left(\frac{8.5}{60}\right)^2 \approx 1240 \text{ miles}$$

or a little less than 2000 kilometres.

S-12: We know that the acceleration of the ball will be constant. If the height of the ball is given by $h(t)$ while it is in the air, $h''(t) = -9.8$. (The negative indicates that the *velocity is decreasing*: the ball starts at its largest velocity, moving in the positive direction, then the velocity decreases to zero and then to a negative number as the ball falls.) As in Example 3.1.2, we need a function $h(t)$ with $h''(t) = -9.8$. Since this is a constant, $h'(t)$ is a line with slope -9.8 , so it has the form

$$h'(t) = -9.8t + a$$

for some constant a . Notice when $t = 0$, $h'(0) = a$, so in fact a is the initial velocity of the ball—the quantity we want to solve for.

Again, as in Example 3.1.2, we need a function $h(t)$ with $h'(t) = -9.8t + a$. Such a function must have the form

$$h(t) = -4.9t^2 + at + b$$

for some constant b . You can find this by guessing and checking, or simply remember it from the text. (In Section 4.1, you'll learn more about figuring out which functions have a particular derivative.) Notice when $t = 0$, $h(0) = b$, so b is the initial height of the baseball, which is 0.

So, $h(t) = -4.9t^2 + at = t(-4.9t + a)$. The baseball is at height zero when it is pitched ($t = 0$) and when it hits the ground (which we want to be $t = 10$). So, we want $(-4.9)(10) + a = 0$. That is, $a = 49$. So, the initial pitch should be at 49 metres per second.

Incidentally, this is on par with the fastest pitch in baseball, as recorded by Guinness World Records:

[https://www.guinnessworldrecords.com/world-records/
fastest-baseball-pitch-\(male\)](https://www.guinnessworldrecords.com/world-records/fastest-baseball-pitch-(male))

S-13: The acceleration of a falling object due to gravity is 9.8 metres per second squared. So, the object's velocity t seconds after being dropped is

$$v(t) = 9.8t \frac{\text{m}}{\text{s}}$$

We want $v(t)$ to be the speed of the peregrine's dive, so we should convert that to metres per second:

$$325 \frac{\text{km}}{\text{hr}} \cdot \left(\frac{1000 \text{ m}}{1 \text{ km}} \right) \left(\frac{1 \text{ hr}}{3600 \text{ sec}} \right) = \frac{1625}{18} \frac{\text{m}}{\text{s}}$$

The stone will reach this velocity when

$$9.8t = \frac{1625}{18} \quad \Rightarrow \quad t = \frac{1625}{18(9.8)}$$

What is left to figure out is how far the stone will fall in this time. The position of the stone $s(t)$ has derivative $9.8t$, so

$$s(t) = 4.9t^2$$

if we take $s(0) = 0$. So, if the stone falls for $\frac{1625}{18(9.8)}$ seconds, in that time it travels

$$s\left(\frac{1625}{18(9.8)}\right) = 4.9 \left(\frac{1625}{18(9.8)}\right)^2 \approx 416 \text{ m}$$

So, you would have to drop a stone from about 416 metres for it to fall as fast as the falcon.

S-14: Since gravity alone brings your cannon ball down, its acceleration is a constant $-9.8 \frac{\text{m}}{\text{s}^2}$. So, $v(t) = v_0 - 9.8t$ and thus its height is given by $s(t) = v_0t - 4.9t^2$ (if we set $s(0) = 0$).

We want to know what value of v_0 makes the maximum height 100 metres. The maximum height is reached when $v(t) = 0$, which is at time $t = \frac{v_0}{9.8}$. So, we solve:

$$\begin{aligned} 100 &= s\left(\frac{v_0}{9.8}\right) \\ 100 &= v_0 \left(\frac{v_0}{9.8}\right) - 4.9 \left(\frac{v_0}{9.8}\right)^2 \\ 100 &= \left(\frac{1}{9.8} - \frac{4.9}{9.8^2}\right) v_0^2 \\ 100 &= \frac{1}{2 \cdot 9.8} v_0^2 \\ v_0^2 &= 1960 \\ v_0 &= \sqrt{1960} \approx 44 \frac{\text{m}}{\text{s}} \end{aligned}$$

where we choose the positive square root because v_0 must be positive for the cannon ball to get off the ground.

S-15: The derivative of acceleration is constant, so the acceleration $a(t)$ has the form $mt + b$. We know $a(0) = -50\,000$ and $a\left(\frac{3}{3600}\right) = -60\,000$ (where we note that 3 seconds is $\frac{3}{3600}$ hours). So, the slope of $a(t)$ is $\frac{-60\,000 + 50\,000}{\frac{3}{3600}} = -12\,000\,000$, which leads us to

$$a(t) = -50\,000 - (12\,000\,000)t$$

where t is measured in hours.

Since $v'(t) = a(t) = -50\,000 - (12\,000\,000)t$, we see

$$v(t) = \frac{-12\,000\,000}{2}t^2 - 50\,000t + c = -6\,000\,000t^2 - 50\,000t + c$$

for some constant c . Since $120 = v(0) = c$:

$$v(t) = -6\,000\,000t^2 - 50\,000t + 120$$

Then after three seconds of braking,

$$\begin{aligned} v\left(\frac{3}{3600}\right) &= -6\,000\,000\left(\frac{3}{3600}\right)^2 - 50\,000\left(\frac{3}{3600}\right) + 120 \\ &= -\frac{25}{6} - \frac{125}{3} + 120 \\ &\approx 74.2 \text{ kph} \end{aligned}$$

Remark: When acceleration is constant, the position function is a quadratic function, but we don't want you to get the idea that position functions are *always* quadratic functions—in the example you just did, it was the velocity function that was quadratic. Position, velocity, and acceleration functions don't have to be polynomial at all—it's only in this section, where we're dealing with the simplest cases, that they seem that way.

S-16: Different forces are acting on you (1) after you jump but before you land on the trampoline, and (2) while you are falling into the trampoline. In both instances, the acceleration is constant, so both height functions are quadratic, of the form $\frac{a}{2}t^2 + vt + h$, where a is the acceleration, v is the velocity when $t = 0$, and h is the initial height.

- Let's consider (1) first, the time during your jump before your feet touch the trampoline. Let $t = 0$ be the moment you jump, and let the rim of the trampoline be height 0. Then, since your initial velocity was (positive) 1 meter per second, your height is given by

$$h_1(t) = \frac{-9.8}{2}t^2 + t = t\left(\frac{-9.8}{2}t + 1\right)$$

Notice that, because your acceleration is working against your positive velocity, it has a negative sign.

- We'll need to know your velocity when your feet first touch the trampoline on your fall. The time your feet first touch the trampoline after your jump is precisely when $h_1(t) = 0$ and $t > 0$. That is, when $t = \frac{2}{9.8}$. Now, since $h'(t) = -9.8t + 1$, $h'\left(\frac{2}{9.8}\right) = -9.8\left(\frac{2}{9.8}\right) + 1 = -1$. So, you are descending at a rate of 1 metre per second at the instant your feet touch the trampoline.

Remark: it is not only coincidence that this was your initial speed. Think about the symmetries of parabolas, and conservation of energy.

- Now we need to think about your height as the trampoline is slowing your fall. One thing to remember about our general equation $\frac{a}{2}t^2 + vt + h$ is that v is the velocity when $t = 0$. But, you don't hit the trampoline at $t = 0$, you hit it at $t = \frac{2}{9.8}$. In order to keep things

simple, let's use a *different* time scale for this second part of your journey. Let's let $h_2(T)$ be your height at time T , from the moment your feet touch the trampoline skin ($T = 0$) to the bottom of your fall. Now, we can use the fact that your initial velocity is -1 metres per second (negative, since your height is decreasing) and your acceleration is 4.9 metres per second per second (positive, since your velocity is increasing from a negative number to zero):

$$h_2(T) = \frac{4.9}{2}T^2 - T$$

where still the height of the rim of the trampoline is taken to be zero.

Remark: if it seems very confusing that your free-falling acceleration is negative, while your acceleration in the trampoline is positive, remember that gravity is pushing you down, but the trampoline is pushing you up.

- How long were you falling in the trampoline? The equation $h_2(T)$ tells you your height only as long as the trampoline is slowing your fall. You reach the bottom of your fall when your velocity is zero.

$$h_2'(T) = 4.9T - 1$$

so you reach the bottom of your fall at $T = \frac{1}{4.9}$. Be careful: this is $\frac{1}{4.9}$ seconds *after you entered the trampoline*, not after the peak of your fall, or after you jumped.

- The last piece of the puzzle is how long it took you to fall from the peak of your jump to the surface of the trampoline. We know the equation of your motion during that time: $h_1(t) = -\frac{9.8}{2}t^2 + t$. You reached the peak when your velocity was zero:

$$h_1'(t) = -9.8t + 1 = 0 \quad \Rightarrow \quad t = \frac{1}{9.8}$$

So, you fell from your peak at $t = \frac{1}{9.8}$ and reached the level of the trampoline rim at $t = \frac{2}{1.98}$, which means the fall took $\frac{1}{9.8}$ seconds.

Remark: by the symmetry mentioned early, the time it took to fall from the peak of your jump to the surface of the trampoline is the same as one-half the time from the moment you jumped off the rim to the moment you're back on the surface of the trampoline.

- So, your time falling from the peak of your jump to its bottom was $\frac{1}{9.8} + \frac{1}{4.9} \approx 0.3$ seconds.

S-17: Let $v(t)$ be the velocity of the object. From the given information:

- $v(0)$ is some value, call it v_0 ,
- $v(1) = 2v_0$ (since the speed doubled in the first second),
- $v(2) = 2(2)v_0$ (since the speed doubled in the second second),
- $v(3) = 2(2)(2)v_0$, and so on.

So, for general t :

$$v(t) = 2^t v(0)$$

To find its acceleration, we simply differentiate. Recall $\frac{d}{dx}\{2^x\} = 2^x \log 2$, where $\log$ denotes logarithm base e .

$$a(t) = 2^t v_0 \log 2$$

Remark: we can also write $a(t) = v(t) \log 2$. The acceleration doubles every second as well.

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S-1: We have an equation relating P and Q :

$$P = Q^3$$

We differentiate implicitly with respect to a third variable, t :

$$\frac{dP}{dt} = 3Q^2 \cdot \frac{dQ}{dt}$$

If we know two of the three quantities $\frac{dP}{dt}$, Q , and $\frac{dQ}{dt}$, then we can find the third. Therefore, ii is a question we can solve. If we know P , then we also know Q (it's just the cube root of P), so also we can solve iv. However, if we know neither P nor Q , then we can't find $\frac{dP}{dt}$ based only off $\frac{dQ}{dt}$, and we can't find $\frac{dQ}{dt}$ based only off $\frac{dP}{dt}$. So we can't solve i or iii.

S-2: Suppose that at time t , the point is at $(x(t), y(t))$. Then $x(t)^2 + y(t)^2 = 1$ so that $2x(t)x'(t) + 2y(t)y'(t) = 0$. We are told that at some time t_0 , $x(t_0) = 2/\sqrt{5}$, $y(t_0) = 1/\sqrt{5}$ and $y'(t_0) = 3$. Then

$$\begin{aligned} 2x(t_0)x'(t_0) + 2y(t_0)y'(t_0) &= 0 &\Rightarrow \\ 2\left(\frac{2}{\sqrt{5}}\right)x'(t_0) + 2\left(\frac{1}{\sqrt{5}}\right)(3) &= 0 &\Rightarrow \\ x'(t_0) &= -\frac{3}{2} \end{aligned}$$

S-3: The instantaneous percentage rate of change for R is

$$\begin{aligned} 100 \frac{R'}{R} &= 100 \frac{(PQ)'}{PQ} && R=PQ \\ &= 100 \frac{P'Q + PQ'}{PQ} && \text{product rule} \\ &= 100 \left[\frac{P'}{P} + \frac{Q'}{Q} \right] && \text{simplify} \\ &= 100[0.08 - 0.02] = 6\% \end{aligned}$$

S-4: (a) By the quotient rule, $F' = \frac{P'Q - PQ'}{Q^2}$. At the moment in question, $F' = \frac{5 \times 5 - 25 \times 1}{5^2} = 0$.

(b) We are told that, at the second moment in time, $P' = 0.1P$ and $Q' = -0.05Q$ (or equivalently $100 \frac{P'}{P} = 10$ and $100 \frac{Q'}{Q} = -5$). Substituting in these values:

$$\begin{aligned}
F' &= \frac{P'Q - PQ'}{Q^2} \\
&= \frac{0.1PQ - P(-0.05Q)}{Q^2} \\
&= \frac{0.15PQ}{Q^2} \\
&= 0.15 \frac{P}{Q} \\
&= 0.15F \quad \implies \\
F' &= 0.15F
\end{aligned}$$

or $100 \frac{F'}{F} = 15\%$. That is, the instantaneous percentage rate of change of F is 15%.

S-5:

- The distance $z(t)$ between the particles at any moment in time is

$$z^2(t) = x(t)^2 + y(t)^2,$$

where $x(t)$ is the position on the x -axis of the particle A at time t (measured in seconds) and $y(t)$ is the position on the y -axis of the particle B at the same time t .

- We differentiate the above equation with respect to t and get

$$2z \cdot z' = 2x \cdot x' + 2y \cdot y',$$

- We are told that $x' = -2$ and $y' = -3$. (The values are negative because x and y are decreasing.) It will take 3 seconds for particle A to reach $x = 4$, and in this time particle B will reach $y = 3$.
- At this point $z = \sqrt{x^2 + y^2} = \sqrt{3^2 + 4^2} = 5$.
- Hence

$$\begin{aligned}
10z' &= 8 \cdot (-2) + 6 \cdot (-3) = -34 \\
z' &= -\frac{34}{10} = -\frac{17}{5} \text{ units per second.}
\end{aligned}$$

S-6:

- We compute the distance $z(t)$ between the two particles after t seconds as

$$z^2(t) = 3^2 + (y_A(t) - y_B(t))^2,$$

where $y_A(t)$ and $y_B(t)$ are the y -coordinates of particles A and B after t seconds, and the horizontal distance between the two particles is always 3 units.

- We are told the distance between the particles is 5 units, this happens when

$$(y_A - y_B)^2 = 5^2 - 3^2 = 16$$

$$y_A - y_B = 4$$

That is, when the difference in y -coordinates is 4. This happens when $t = 4$.

- We differentiate the distance equation (from the first bullet point) with respect to t and get

$$2z \cdot z' = 2(y_A' - y_B')(y_A - y_B),$$

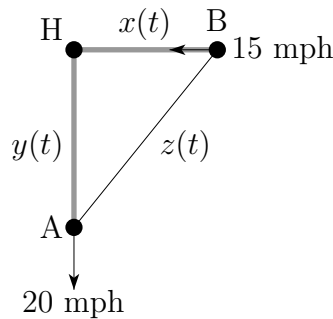
- We know that $(y_A - y_B) = 4$, and we are told that $z = 5$, $y_A' = 3$, and $y_B' = 2$. Hence

$$10z'(4) = 2 \times 1 \times 4 = 8$$

- Therefore

$$z'(4) = \frac{8}{10} = \frac{4}{5} \text{ units per second.}$$

S-7:



As in the above figure, let $x(t)$ be the distance between H (Hawaii) and ship B, and $y(t)$ be the distance between H and ship A, and $z(t)$ be the distance between ships A and B, all at time t . Then

$$x(t)^2 + y(t)^2 = z(t)^2$$

Differentiating with respect to t ,

$$2x(t)x'(t) + 2y(t)y'(t) = 2z(t)z'(t)$$

$$x(t)x'(t) + y(t)y'(t) = z(t)z'(t)$$

At the specified time, $x(t)$ is decreasing, so $x'(t)$ is negative, and $y(t)$ is increasing, so $y'(t)$ is positive.

$$(300)(-15) + (400)(20) = \sqrt{300^2 + 400^2}z'(t)$$

$$500z'(t) = 3500$$

$$z'(t) = 7 \text{ mph}$$

S-8:

-
- We compute the distance $d(t)$ between the two snails after t minutes as

$$d^2(t) = 30^2 + (y_1(t) - y_2(t))^2,$$

where $y_1(t)$ is the altitude of the first snail, and $y_2(t)$ the altitude of the second snail after t minutes.

- We differentiate the above equation with respect to t and get

$$2d \cdot d' = 2(y_1' - y_2')(y_1 - y_2)$$

$$d \cdot d' = (y_1' - y_2')(y_1 - y_2)$$

- We are told that $y_1' = 25$ and $y_2' = 15$. It will take 4 minutes for the first snail to reach $y_1 = 100$, and in this time the second snail will reach $y_2 = 60$.
- At this point $d^2 = 30^2 + (100 - 60)^2 = 900 + 1600 = 2500$, hence $d = 50$.
- Therefore

$$50d' = (25 - 15) \times (100 - 60)$$

$$d' = \frac{400}{50} = 8 \text{ cm per minute.}$$

S-9:

- If we write $z(t)$ for the length of the ladder at time t and $y(t)$ for the height of the top end of the ladder at time t we have

$$z(t)^2 = 5^2 + y(t)^2.$$

- We differentiate the above equation with respect to t and get

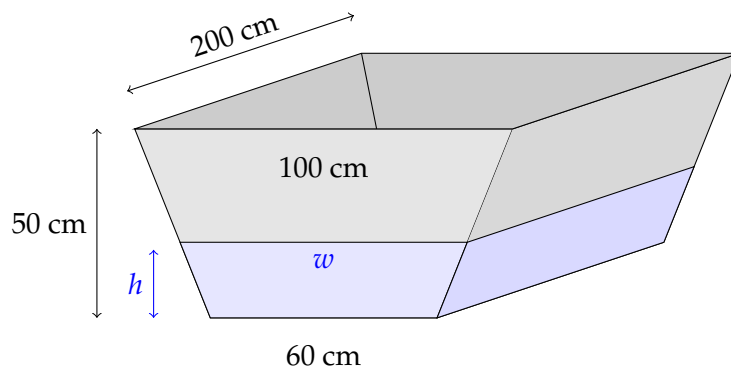
$$2z \cdot z' = 2y \cdot y',$$

- We are told that $z'(t) = -2$, so $z(3.5) = 20 - 3.5 \cdot 2 = 13$.
- At this point $y = \sqrt{z^2 - 5^2} = \sqrt{169 - 25} = \sqrt{144} = 12$.
- Hence

$$2 \cdot 13 \cdot (-2) = 2 \cdot 12y'$$

$$y' = -\frac{2 \cdot 13}{12} = -\frac{13}{6} \text{ meters per second.}$$

S-10: What we're given is $\frac{dV}{dt}$ (where V is volume of water in the trough, and t is time), and what we are asked for is $\frac{dh}{dt}$ (where h is the height of the water). So, we need an equation relating V and h . First, let's get everything in the same units: centimetres.



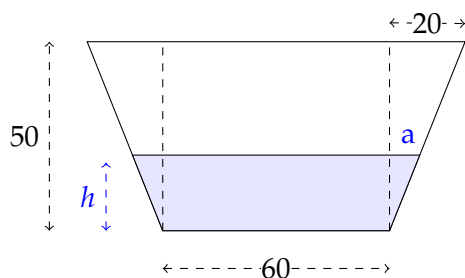
We can calculate the volume of water in the trough by multiplying the area of its trapezoidal cross section by 200 cm. A trapezoid with height h and bases b_1 and b_2 has area $h \left(\frac{b_1 + b_2}{2} \right)$. (To see why this is so, draw the trapezoid as a rectangle flanked by two triangles.) So, using w as the width of the top of the water (as in the diagram above), the area of the cross section of the water in the trough is

$$A = h \left(\frac{60 + w}{2} \right)$$

and therefore the volume of water in the trough is

$$V = 100h(60 + w) \text{ cm}^3.$$

We need a formula for w in terms of h . If we draw lines straight up from the bottom corners of the trapezoid, we break it into rectangles and triangles.



Using similar triangles, $\frac{a}{h} = \frac{20}{50}$, so $a = \frac{2}{5}h$. Then

$$\begin{aligned} w &= 60 + 2a \\ &= 60 + 2 \left(\frac{2}{5}h \right) = 60 + \frac{4}{5}h \end{aligned}$$

so

$$\begin{aligned} V &= 100h(60 + w) \\ &= 100h(120 + \frac{4}{5}h) \\ &= 80h^2 + 12000h \end{aligned}$$

This is the equation we need, relating V and h . Differentiating implicitly with respect to t :

$$\begin{aligned} \frac{dV}{dt} &= 2 \cdot 80h \cdot \frac{dh}{dt} + 12000 \frac{dh}{dt} \\ &= (160h + 12000) \frac{dh}{dt} \end{aligned}$$

We are given that $h = 25$ and $\frac{dV}{dt} = 3$ litres per minute. Converting to cubic centimetres, $\frac{dV}{dt} = -3000$ cubic centimetres per minute. So:

$$\begin{aligned} -3000 &= (160 \cdot 25 + 12000) \frac{dh}{dt} \\ \frac{dh}{dt} &= -\frac{3}{16} = -.1875 \frac{\text{cm}}{\text{min}} \end{aligned}$$

So, the water level is dropping at $\frac{3}{16}$ centimetres per minute.

S-11: If V is the volume of the water in the tank, and t is time, then we are given $\frac{dV}{dt}$. What we want to know is $\frac{dh}{dt}$, where h is the height of the water in the tank. A reasonable plan is to find an equation relating V and h , and differentiate it implicitly with respect to t .

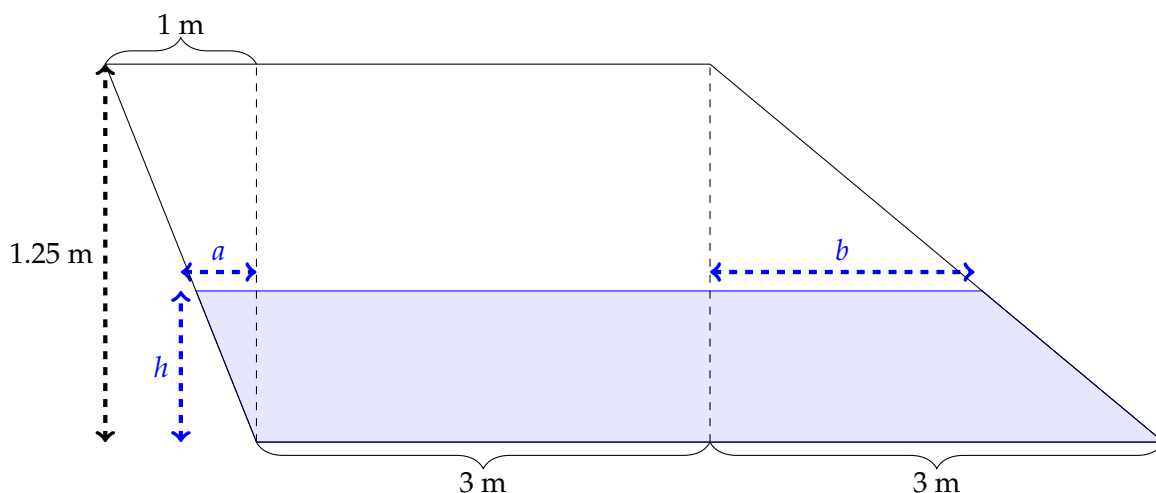
Let's be a little careful about units. The volume of water in the tank is

$$(\text{area of cross section of water}) \times (\text{length of tank})$$

If we measure these values in metres (area in square metres, length in metres), then the volume is going to be in cubic metres. So, when we differentiate with respect to time, our units will be cubic metres per second. The water is flowing in at one litre per second, or 1000 cubic centimetres per second. So, we either have to measure our areas and distances in centimetres, or convert litres to cubic metres. We'll do the latter, but both are fine.

If we imagine one cubic metre as a cube, with each side of length 1 metre, then it's easy to see the volume inside is $(100)^3 = 10^6$ cubic centimetres: it's the volume of a cube with each side of length 100 cm. Since a litre is 10^3 cubic centimetres, and a cubic metre is 10^6 cubic centimetres, one litre is 10^{-3} cubic metres. So, $\frac{dV}{dt} = \frac{1}{10^3}$ cubic metres per second.

Let h be the height of the water (in metres). We can figure out the area of the cross section by breaking it into three pieces: a triangle on the left, a rectangle in the middle, and a trapezoid on the right.



- The triangle on the left has height h metres. Let its base be a metres. It forms a similar triangle with the triangle whose height is 1.25 metres and width is 1 metre, so:

$$\frac{a}{h} = \frac{1}{1.25}$$

$$a = \frac{4}{5}h$$

So, the area of the triangle on the left is

$$\frac{1}{2}ah = \frac{2}{5}h^2$$

- The rectangle in the middle has length 3 metres and height h metres, so its area is $3h$ square metres.
- The trapezoid on the right is a portion of a triangle with base 3 metres and height 1.25 metres. So, its area is

$$\underbrace{\left(\frac{1}{2}(3)(1.25)\right)}_{\text{area of big triangle}} - \underbrace{\left(\frac{1}{2}(b)(1.25 - h)\right)}_{\text{area of little triangle}}$$

The little triangle (of base b and height $1.25 - h$) is formed by the *air* on the right side of the tank. It is a similar triangle to the triangle of base 3 and height 1.25, so

$$\frac{b}{1.25 - h} = \frac{3}{1.25}$$

$$b = \frac{3}{1.25}(1.25 - h)$$

So, the area of the trapezoid on the right is

$$\frac{1}{2}(3)(1.25) - \frac{1}{2}\left(\frac{3}{1.25}\right)(1.25 - h)(1.25 - h)$$

$$= 3h - \frac{6}{5}h^2$$

So, the area A of the cross section of the water is

$$A = \underbrace{\frac{2}{5}h^2}_{\text{triangle}} + \underbrace{3h}_{\text{rectangle}} + \underbrace{3h - \frac{6}{5}h^2}_{\text{trapezoid}}$$

$$= 6h - \frac{4}{5}h^2$$

So, the volume of water is

$$V = 5\left(6h - \frac{4}{5}h^2\right) = 30h - 4h^2$$

Differentiating with respect to time, t :

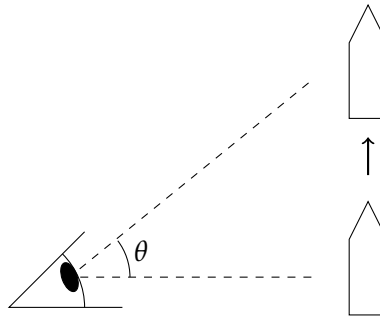
$$\frac{dV}{dt} = 30\frac{dh}{dt} - 8h\frac{dh}{dt}$$

When $h = \frac{1}{10}$ metre, and $\frac{dV}{dt} = \frac{1}{10^3}$ cubic metres per second,

$$\begin{aligned}\frac{1}{10^3} &= 30 \frac{dh}{dt} - 8 \left(\frac{1}{10} \right) \frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{1}{29200} \text{ metres per second}\end{aligned}$$

This is about 1 centimetre every five minutes. You might want a bigger hose.

S-12: Let θ be the angle of your head, where $\theta = 0$ means you are looking straight ahead, and $\theta = \frac{\pi}{2}$ means you are looking straight up. We are interested in $\frac{d\theta}{dt}$, but we only have information about h . So, a reasonable plan is to find an equation relating h and θ , and differentiate with respect to time.



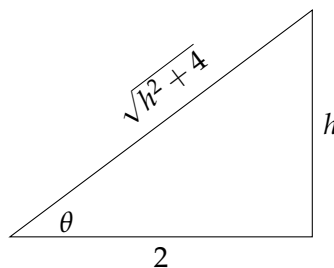
The right triangle formed by you, the rocket, and the rocket's original position has adjacent side (to θ) length 2km, and opposite side (to θ) length $h(t)$ kilometres, so

$$\tan \theta = \frac{h}{2}$$

Differentiating with respect to t :

$$\begin{aligned}\sec^2 \theta \cdot \frac{d\theta}{dt} &= \frac{1}{2} \frac{dh}{dt} \\ \frac{d\theta}{dt} &= \frac{1}{2} \cos^2 \theta \cdot \frac{dh}{dt}\end{aligned}$$

We know $\tan \theta = \frac{h}{2}$. We draw a right triangle with angle θ (filling in the sides using SOH CAH TOA and the Pythagorean theorem) to figure out $\cos \theta$:



Using the triangle, $\cos \theta = \frac{2}{\sqrt{h^2 + 4}}$, so

$$\begin{aligned}\frac{d\theta}{dt} &= \frac{1}{2} \left(\frac{2}{\sqrt{h^2 + 4}} \right)^2 \cdot \frac{dh}{dt} \\ &= \left(\frac{2}{h^2 + 4} \right) \frac{dh}{dt}\end{aligned}$$

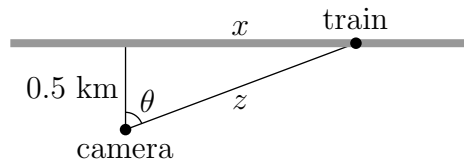
So, the quantities we need to know one minute after liftoff (that is, when $t = \frac{1}{60}$) are $h\left(\frac{1}{60}\right)$ and $\frac{dh}{dt}\left(\frac{1}{60}\right)$. Recall $h(t) = 61750t^2$.

$$\begin{aligned}h\left(\frac{1}{60}\right) &= \frac{61750}{3600} = \frac{1235}{72} \\ \frac{dh}{dt} &= 2(61750)t \\ \frac{dh}{dt}\left(\frac{1}{60}\right) &= \frac{2(61750)}{60} = \frac{6175}{3}\end{aligned}$$

Returning to the equation $\frac{d\theta}{dt} = \left(\frac{2}{h^2 + 4} \right) \frac{dh}{dt}$:

$$\frac{d\theta}{dt}\left(\frac{1}{60}\right) = \left(\frac{2}{\left(\frac{1235}{72}\right)^2 + 4} \right) \left(\frac{6175}{3} \right) \approx 13.8 \frac{\text{rad}}{\text{hour}} \approx 0.0038 \frac{\text{rad}}{\text{sec}}$$

S-13: (a) Let $x(t)$ be the distance of the train along the track at time t , measured from the point on the track nearest the camera. Let $z(t)$ be the distance from the camera to the train at time t .



Then $x'(t) = 2$ and at the time in question, $z(t) = 1.3$ km and $x(t) = \sqrt{1.3^2 - 0.5^2} = 1.2$ km. So

$$\begin{aligned}z(t)^2 &= x(t)^2 + 0.5^2 \\ 2z(t)z'(t) &= 2x(t)x'(t) \\ 2 \times 1.3z'(t) &= 2 \times 1.2 \times 2 \\ z'(t) &= \frac{2 \times 1.2}{1.3} \approx 1.85 \text{ km/min}\end{aligned}$$

(b) Let $\theta(t)$ be the angle shown at time t . Then

$$\sin(\theta(t)) = \frac{x(t)}{z(t)}$$

Differentiating with respect to t :

$$\begin{aligned}\theta'(t) \cos(\theta(t)) &= \frac{x'(t)z(t) - x(t)z'(t)}{z(t)^2} \\ \theta'(t) &= \frac{x'(t)z(t) - x(t)z'(t)}{z(t)^2 \cos(\theta(t))}\end{aligned}$$

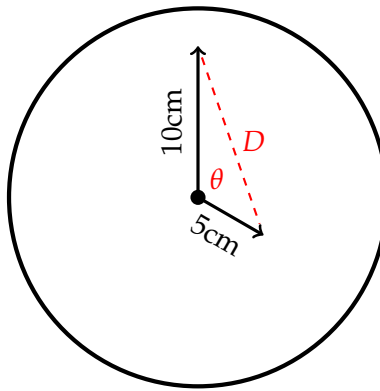
From our diagram, we see $\cos(\theta(t)) = \frac{0.5}{z(t)}$, so:

$$= 2 \frac{x'(t)z(t) - x(t)z'(t)}{z(t)}$$

Substituting in $x'(t) = 2$, $z(t) = 1.3$, $x(t) = 1.2$, and $z'(t) = \frac{2 \times 1.2}{1.3}$:

$$\theta'(t) = 2 \frac{2 \times 1.3 - 1.2 \times \frac{2 \times 1.2}{1.3}}{1.3} \approx .592 \text{ radians/min}$$

S-14: Let θ be the angle between the two hands.



The Law of Cosines (Appendix B.4.1) tells us that

$$\begin{aligned} D^2 &= 5^2 + 10^2 - 2 \cdot 5 \cdot 10 \cdot \cos \theta \\ D^2 &= 125 - 100 \cos \theta \end{aligned}$$

Differentiating with respect to time t ,

$$2D \frac{dD}{dt} = 100 \sin \theta \cdot \frac{d\theta}{dt}$$

Our tasks now are to find D , θ and $\frac{d\theta}{dt}$ when the time is 4:00. At 4:00, the minute hand is straight up, and the hour hand is $\frac{4}{12} = \frac{1}{3}$ of the way around the clock, so $\theta = \frac{1}{3}(2\pi) = \frac{2\pi}{3}$ at 4:00. Then $D^2 = 125 - 100 \cos(\frac{2\pi}{3}) = 125 - 100(-\frac{1}{2}) = 175$, so $D = \sqrt{175} = 5\sqrt{7}$ at 4:00.

To calculate $\frac{d\theta}{dt}$, remember that both hands are moving. The hour hand makes a full rotation every 12 hours, so its rotational speed is $\frac{2\pi}{12} = \frac{\pi}{6}$ radians per hour. The hour hand is being chased by the minute hand. The minute hand makes a full rotation every hour, so its rotational speed is $\frac{2\pi}{1} = 2\pi$ radians per hour. Therefore, the angle θ between the two hands is changing at a rate of

$$\frac{d\theta}{dt} = -\left(2\pi - \frac{\pi}{6}\right) = -\frac{11\pi}{6} \frac{\text{rad}}{\text{hr}}.$$

Now, we plug in D , θ , and $\frac{d\theta}{dt}$ to find $\frac{dD}{dt}$:

$$\begin{aligned} 2D \frac{dD}{dt} &= 100 \sin \theta \cdot \frac{d\theta}{dt} \\ 2(5\sqrt{7}) \frac{dD}{dt} &= 100 \sin\left(\frac{2\pi}{3}\right) \left(\frac{-11\pi}{6}\right) \\ 10\sqrt{7} \frac{dD}{dt} &= 100 \left(\frac{\sqrt{3}}{2}\right) \left(\frac{-11\pi}{6}\right) = -\frac{275\pi}{\sqrt{3}} \\ \frac{dD}{dt} &= \frac{-55\sqrt{21}\pi}{42} \frac{\text{cm}}{\text{hr}} \end{aligned}$$

So D is decreasing at $\frac{55\sqrt{21}\pi}{42} \approx 19$ centimetres per hour.

S-15: The area at time t is the area of the outer circle minus the area of the inner circle:

$$\begin{aligned} A(t) &= \pi(R(t)^2 - r(t)^2) \\ \text{So, } A'(t) &= 2\pi(R(t)R'(t) - r(t)r'(t)) \end{aligned}$$

Plugging in the given data,

$$A' = 2\pi(3 \cdot 2 - 1 \cdot 7) = -2\pi$$

So the area is *shrinking* at a rate of $2\pi \frac{\text{cm}^2}{\text{s}}$.

S-16: The volume between the spheres, while the little one is inside the big one, is

$$V = \frac{4}{3}\pi R^3 - \frac{4}{3}\pi r^3$$

Differentiating implicitly with respect to t :

$$\frac{dV}{dt} = 4\pi R^2 \frac{dR}{dt} - 4\pi r^2 \frac{dr}{dt}$$

We differentiate $R = 10 + 2t$ and $r = 6t$ to find $\frac{dR}{dt} = 2$ and $\frac{dr}{dt} = 6$. When $R = 2r$, $10 + 2t = 2(6t)$, so $t = 1$. When $t = 1$, $R = 12$ and $r = 6$. So:

$$\frac{dV}{dt} = 4\pi (12^2) (2) - 4\pi (6^2) (6) = 288\pi$$

So the volume between the two spheres is increasing at 288π cubic units per unit time.

Remark: when the radius of the inner sphere increases, we are “subtracting” more area. Since the radius of the inner sphere grows faster than the radius of the outer sphere, we might expect the area between the spheres to be decreasing. Although the radius of the outer sphere grows more slowly, a small increase in the radius of the outer sphere results in a larger change in volume than the same increase in the radius of the inner sphere. So, a result showing that the volume between the spheres is increasing is not unreasonable.

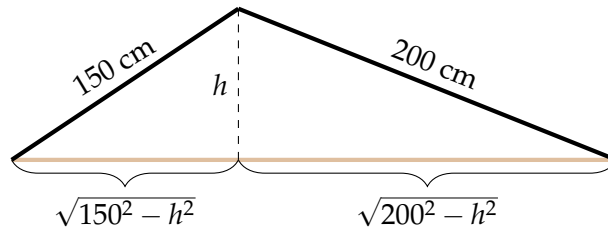
S-17: We know something about the rate of change of the height h of the triangle, and we want to know something about the rate of change of its area, A . A reasonable plan is to find an equation relating A and h , and differentiate implicitly with respect to t . The area of a triangle with height h and base b is

$$A = \frac{1}{2}bh$$

Note, b will change with time as well as h . So, differentiating with respect to time, t :

$$\frac{dA}{dt} = \frac{1}{2} \left(\frac{db}{dt} \cdot h + b \cdot \frac{dh}{dt} \right)$$

We are given $\frac{dh}{dt}$ and h , but those b 's are a mystery. We need to relate them to h . We can do this by breaking our triangle into two right triangles and using the Pythagorean Theorem:



So, the base of the triangle is

$$b = \sqrt{150^2 - h^2} + \sqrt{200^2 - h^2}$$

Differentiating with respect to t :

$$\begin{aligned} \frac{db}{dt} &= \frac{-2h \frac{dh}{dt}}{2\sqrt{150^2 - h^2}} + \frac{-2h \frac{dh}{dt}}{2\sqrt{200^2 - h^2}} \\ &= \frac{-h \frac{dh}{dt}}{\sqrt{150^2 - h^2}} + \frac{-h \frac{dh}{dt}}{\sqrt{200^2 - h^2}} \end{aligned}$$

Using $\frac{dh}{dt} = -3$ centimetres per minute:

$$\frac{db}{dt} = \frac{3h}{\sqrt{150^2 - h^2}} + \frac{3h}{\sqrt{200^2 - h^2}}$$

When $h = 120$, $\sqrt{150^2 - h^2} = 90$ and $\sqrt{200^2 - h^2} = 160$. So, at this moment in time:

$$\begin{aligned} b &= 90 + 160 = 250 \\ \frac{db}{dt} &= \frac{3(120)}{90} + \frac{3(120)}{160} = 4 + \frac{9}{4} = \frac{25}{4} \end{aligned}$$

We return to our equation relating the derivatives of A , b , and h .

$$\frac{dA}{dt} = \frac{1}{2} \left(\frac{db}{dt} \cdot h + b \cdot \frac{dh}{dt} \right)$$

When $h = 120$ cm, $b = 250$, $\frac{dh}{dt} = -3$, and $\frac{db}{dt} = \frac{25}{4}$:

$$\begin{aligned} \frac{dA}{dt} &= \frac{1}{2} \left(\frac{25}{4}(120) + 250(-3) \right) \\ &= 0 \end{aligned}$$

Remark: What does it mean that $\left. \frac{dA}{dt} \right|_{h=120} = 0$? Certainly, as the height changes, the area changes as well. As the height sinks to 120 cm, the area is increasing, but after it sinks *past* 120 cm, the area is *decreasing*. So, at the instant when the height is exactly 120 cm, the area is neither increasing nor decreasing: it is at a local maximum. You'll learn more about this kind of problem in Section 3.5.

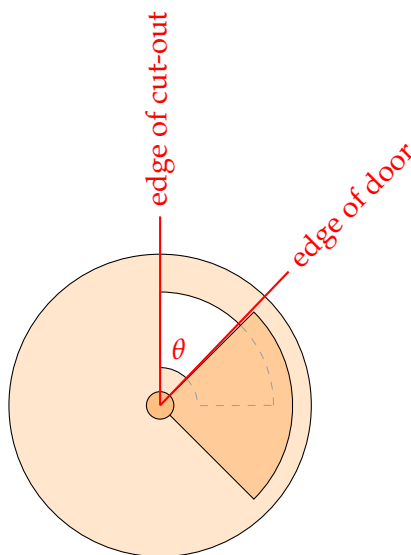
S-18: Let S be the flow of salt (in cubic centimetres per second). We want to know $\frac{dS}{dt}$: how fast the flow is changing at time t . We are given an equation for S :

$$S = \frac{1}{5}A$$

where A is the uncovered area of the cut-out. So,

$$\frac{dS}{dt} = \frac{1}{5} \frac{dA}{dt}$$

If we can find $\frac{dA}{dt}$, then we can find $\frac{dS}{dt}$. We are given information about how quickly the door is rotating. If we let θ be the angle made by the leading edge of the door and the far edge of the cut-out (shown below), then $\frac{d\theta}{dt} = -\frac{\pi}{6}$ radians per second. (Since the door is covering more and more of the cut-out, θ is getting smaller, so $\frac{d\theta}{dt}$ is negative.)



Since we know $\frac{d\theta}{dt}$, and we want to know $\frac{dA}{dt}$ (in order to get $\frac{dS}{dt}$), it is reasonable to look for an equation relating A and θ , and differentiate it implicitly with respect to t to get an equation relating $\frac{dA}{dt}$ and $\frac{d\theta}{dt}$.

The area of an annulus with outer radius 6 cm and inner radius 1 cm is $\pi \cdot 6^2 - \pi \cdot 1^2 = 35\pi$ square centimetres. A sector of that same annulus with angle θ has area $\left(\frac{\theta}{2\pi}\right)(35\pi)$, since $\frac{\theta}{2\pi}$ is the ratio of the sector to the entire annulus. (For example, if $\theta = \pi$, then the sector is *half* of the entire annulus, so its area is $(1/2)35\pi$.)

So, when $0 \leq \theta \leq \frac{\pi}{2}$, the area of the cutout that is open is

$$A = \frac{\theta}{2\pi}(35\pi) = \frac{35}{2}\theta$$

This is the formula we wanted, relating A and θ . Differentiating with respect to t ,

$$\frac{dA}{dt} = \frac{35}{2} \frac{d\theta}{dt} = \frac{35}{2} \left(-\frac{\pi}{6} \right) = -\frac{35\pi}{12}$$

Since $\frac{dS}{dt} = \frac{1}{5} \frac{dA}{dt}$,

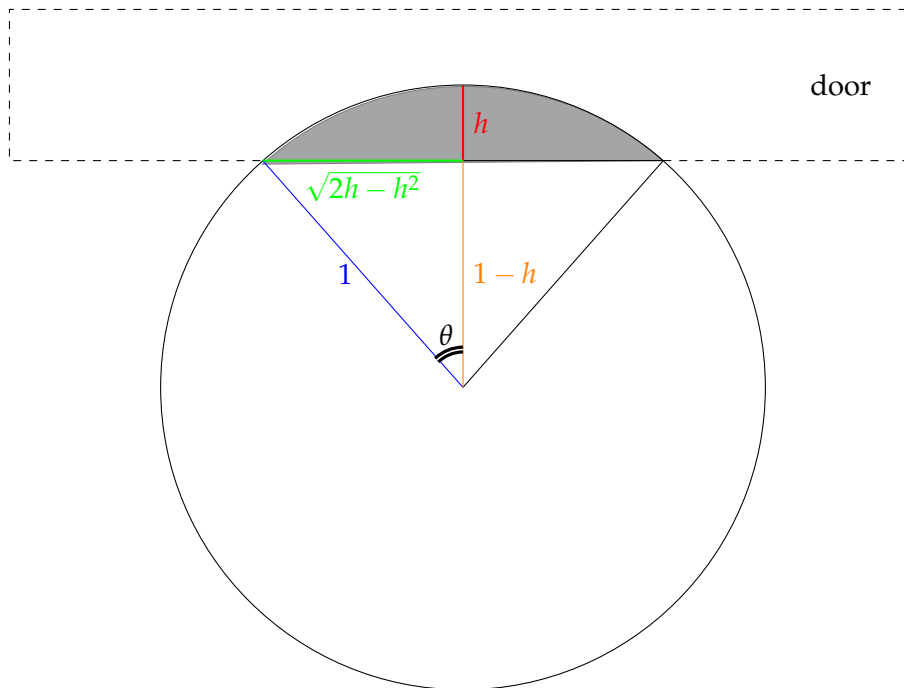
$$\frac{dS}{dt} = -\frac{1}{5} \frac{35\pi}{12} = -\frac{7\pi}{12} \approx -1.8 \frac{\text{cm}^3}{\text{sec}^2}$$

Remark: the change in flow of salt is constant while the door covers more and more of the cut-out, so we never used the fact that precisely half of the cut-out was open. We also never used the radius of the lid, which is immaterial to the flow of salt.

S-19: Let F be the flow of water through the pipe, so $F = \frac{1}{5}A$. We want to know $\frac{dF}{dt}$, so differentiating implicitly with respect to t , we find

$$\frac{dF}{dt} = \frac{1}{5} \frac{dA}{dt}.$$

If we can find $\frac{dA}{dt}$, then we can find $\frac{dF}{dt}$. We know something about the shape of the uncovered area of the pipe; a reasonable plan is to find an equation relating the height of the door with the uncovered area of the pipe. Let h be the distance from the top of the pipe to the bottom of the door, measured in metres.



Since the radius of the pipe is 1 metre, the orange line has length $1-h$ metres, and the blue line has length 1 metre. Using the Pythagorean Theorem, the green line has length $\sqrt{1^2 - (1-h)^2} = \sqrt{2h-h^2}$ metres.

The uncovered area of the pipe can be broken up into a triangle (of height $1 - h$ and base $2\sqrt{2h - h^2}$) and a sector of a circle (with angle $2\pi - 2\theta$). The area of the triangle is

$$\underbrace{(1 - h)}_{\text{height}} \underbrace{\sqrt{2h - h^2}}_{\frac{1}{2}\text{base}}.$$

The area of the sector is

$$\underbrace{\left(\frac{2\pi - 2\theta}{2\pi}\right)}_{\text{fraction of circle}} \underbrace{(\pi \cdot 1^2)}_{\text{area of circle}} = \pi - \theta.$$

Remember: what we want is to find $\frac{dA}{dt}$, and what we know is $\frac{dh}{dt} = 0.01$ metres per second. If we find θ in terms of h , we find A in terms of h , and then differentiate with respect to t .

Since θ is an angle in a right triangle with hypotenuse 1 and adjacent side length $1 - h$, $\cos \theta = \frac{1-h}{1} = 1 - h$. We want to conclude that $\theta = \arccos(1 - h)$, but let's be a little careful: remember that the range of the arccosine function is angles in $[0, \pi]$. We must be confident that $0 \leq \theta \leq \pi$ in order to conclude $\theta = \arccos(1 - h)$ —but clearly, θ is in this range. (Remark: we could also have said $\sin \theta = \frac{\sqrt{2h-h^2}}{1}$, and so $\theta = \arcsin(\sqrt{2h-h^2})$. This would require $-\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$, which is true when $h < 1$, but false for $h > 1$. Since our problem asks about $h = 0.25$, we could also use arcsine.)

Now, we know the area of the open pipe in terms of h .

$$\begin{aligned} A &= (\text{area of triangle}) + (\text{area of sector}) \\ &= (1 - h)\sqrt{2h - h^2} + (\pi - \theta) \\ &= (1 - h)\sqrt{2h - h^2} + \pi - \arccos(1 - h) \end{aligned}$$

We want to differentiate with respect to t . Using the chain rule:

$$\begin{aligned} \frac{dA}{dt} &= \frac{dA}{dh} \cdot \frac{dh}{dt} \\ \frac{dA}{dt} &= \left((1 - h) \frac{2 - 2h}{2\sqrt{2h - h^2}} + (-1)\sqrt{2h - h^2} + \frac{-1}{\sqrt{1 - (1 - h)^2}} \right) \frac{dh}{dt} \\ &= \left(\frac{(1 - h)^2}{\sqrt{2h - h^2}} - \sqrt{2h - h^2} - \frac{1}{\sqrt{2h - h^2}} \right) \frac{dh}{dt} \\ &= \left(\frac{(1 - h)^2 - 1}{\sqrt{2h - h^2}} - \sqrt{2h - h^2} \right) \frac{dh}{dt} \\ &= \left(\frac{-(2h - h^2)}{\sqrt{2h - h^2}} - \sqrt{2h - h^2} \right) \frac{dh}{dt} \\ &= \left(-\sqrt{2h - h^2} - \sqrt{2h - h^2} \right) \frac{dh}{dt} \\ &= -2\sqrt{2h - h^2} \frac{dh}{dt} \end{aligned}$$

We note here that the negative sign makes sense: as the door lowers, h increases and A decreases, so $\frac{dh}{dt}$ and $\frac{dA}{dt}$ should have opposite signs.

When $h = \frac{1}{4}$ metres, and $\frac{dh}{dt} = \frac{1}{100}$ metres per second:

$$\frac{dA}{dt} = -2\sqrt{\frac{2}{4} - \frac{1}{4^2}} \left(\frac{1}{100} \right) = -\frac{\sqrt{7}}{200} \frac{\text{cm}^2}{\text{s}}$$

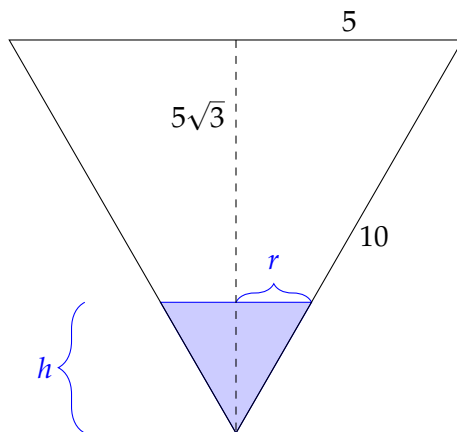
Since $\frac{dF}{dt} = \frac{1}{5} \frac{dA}{dt}$:

$$\frac{dF}{dt} = -\frac{\sqrt{7}}{1000} \frac{\text{m}^3}{\text{sec}^2}$$

That is, the flow is decreasing at a rate of $\frac{\sqrt{7}}{1000} \frac{\text{m}^3}{\text{sec}^2}$.

S-20: We are given the rate of change of the volume of liquid, and are asked for the rate of change of the height of the liquid. So, we need an equation relating volume and height.

The volume V of a cone with height h and radius r is $\frac{1}{3}\pi r^2 h$. Since we know $\frac{dV}{dt}$, and want to know $\frac{dh}{dt}$, we need to find a way to deal with the unwanted variable r . We can find r in terms of h by using similar triangles. Viewed from the side, the conical glass is an equilateral triangle, as is the water in it. Using the Pythagorean Theorem, the cone has height $5\sqrt{3}$.



Using similar triangles, $\frac{r}{h} = \frac{5}{5\sqrt{3}}$, so $r = \frac{h}{\sqrt{3}}$. (Remark: we could also use the fact that the water forms a cone that looks like an equilateral triangle when viewed from the side to conclude $r = \frac{h}{\sqrt{3}}$.)

Now, we can write the volume of water in the cone in terms of h , and no other variables.

$$\begin{aligned} V &= \frac{1}{3}\pi r^2 h \\ &= \frac{1}{3}\pi \left(\frac{h}{\sqrt{3}} \right)^2 h \\ &= \frac{\pi}{9} h^3 \end{aligned}$$

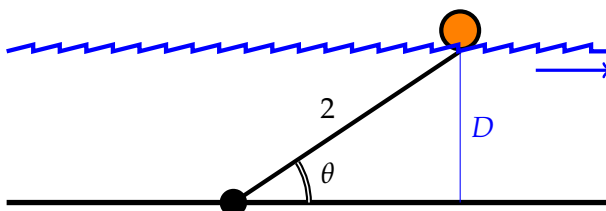
Differentiating with respect to t :

$$\frac{dV}{dt} = \frac{\pi}{3} h^2 \frac{dh}{dt}$$

When $h = 7$ cm and $\frac{dV}{dt} = -5$ mL per minute,

$$\begin{aligned} -5 &= \frac{\pi}{3}(49)\frac{dh}{dt} \\ \frac{dh}{dt} &= \frac{-15}{49\pi} \approx -0.097 \text{ cm per minute} \end{aligned}$$

S-21: As is so often the case, we use a right triangle in this problem to relate the quantities.



$$\begin{aligned} \sin \theta &= \frac{D}{2} \\ D &= 2 \sin \theta \end{aligned}$$

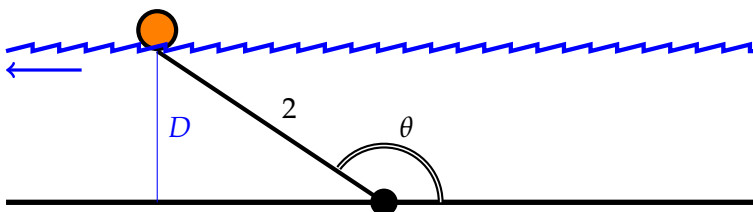
Using the chain rule, we differentiate both sides with respect to time, t .

$$\frac{dD}{dt} = 2 \cos \theta \cdot \frac{d\theta}{dt}$$

So, if $\frac{d\theta}{dt} = 0.25$ radians per hour and $\theta = \frac{\pi}{4}$ radians, then

$$(a) \quad \frac{dD}{dt} = 2 \cos \left(\frac{\pi}{4} \right) \cdot 0.25 = 2 \left(\frac{1}{\sqrt{2}} \right) \frac{1}{4} = \frac{1}{2\sqrt{2}} \text{ metres per hour.}$$

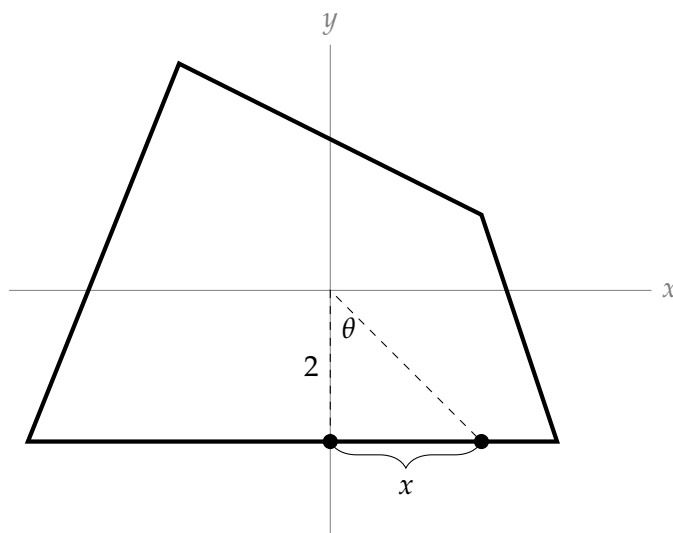
Setting aside part (b) for a moment, let's think about (c). If $\frac{d\theta}{dt}$ and $\frac{dD}{dt}$ have different signs, then because $\frac{dD}{dt} = 2 \cos \theta \cdot \frac{d\theta}{dt}$, that means $\cos \theta < 0$. We have to have a nonnegative depth, so $D > 0$ and $D = 2 \sin \theta$ implies $\sin \theta > 0$. If $\sin \theta \geq 0$ and $\cos \theta < 0$, then $\theta \in (\pi/2, \pi]$. On the diagram, that looks like this:



That is: the water has reversed direction. This happens, for instance, when a river empties into the ocean and the tide is high. Skookumchuck Narrows provincial park, in the Sunshine Coast, has reversing rapids.

Now, let's return to (b). If the rope is only 2 metres long, and the river rises *higher* than 2 metres, then our equation $D = 2 \sin \theta$ doesn't work any more: the buoy might be stationary underwater while the water rises or falls (but stays at or above 2 metres deep).

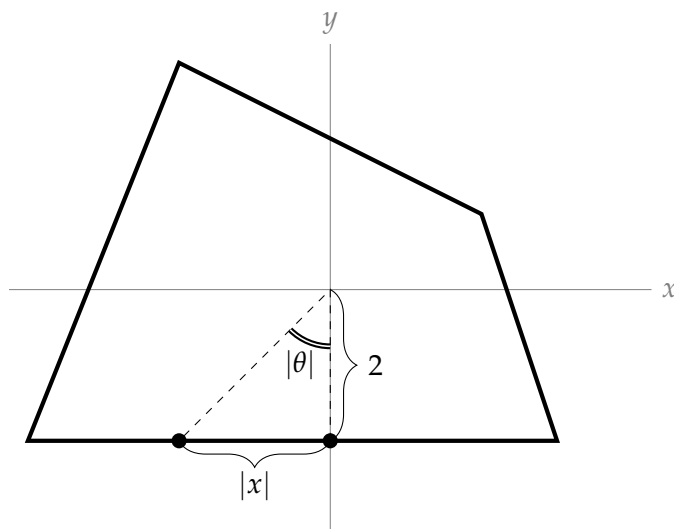
S-22: (a) When the point is at $(0, -2)$, its y -coordinate is not changing, because it is moving along a horizontal line. So, the rate at which the particle moves is simply $\frac{dx}{dt}$. Let θ be the angle an observer would be looking at, in order to watch the point. Since we know $\frac{d\theta}{dt}$, a reasonable plan is to find an equation relating θ and x , and then differentiate implicitly with respect to t . To do this, let's return to our diagram.



When the point is a little to the right of $(0, -2)$, then we can make a triangle with the origin, as shown. If we let θ be the indicated angle, then $\frac{d\theta}{dt} = 1$ radian per second. (It is given that the observer is turning one radian per second, so this is how fast θ is increasing.) From the right triangle in the diagram, we see

$$\tan \theta = \frac{x}{2}$$

Now, we have to take care of a subtle point. The diagram we drew only makes sense for the point when it is at a position a little to the *right* of $(0, -2)$. So, right now, we've only made a set-up that will find the derivative *from the right*. But, with a little more thought, we see that even when x is negative (that is, when the point is a little to the *left* of $(0, -2)$), our equation holds if we are careful about how we define θ . Let θ be the angle between the line connecting the point and the origin, and the y -axis, where θ is *negative* when the point is to the left of the y -axis.



Since x and θ are both negative when the point is to the left of the y -axis,

$$\begin{aligned}\tan |\theta| &= \frac{2}{|x|} \\ \tan(-\theta) &= \frac{-x}{2}\end{aligned}$$

So, since $\tan(-\theta) = -\tan(\theta)$:

$$\tan \theta = \frac{x}{2}$$

So, we've shown that the relationship $\tan \theta = \frac{x}{2}$ holds when our point is at $(x, -2)$, regardless of the sign of x .

Moving on, since we are given $\frac{d\theta}{dt}$ and asked for $\frac{dx}{dt}$, we differentiate with respect to t :

$$\sec^2 \theta \cdot \frac{d\theta}{dt} = \frac{1}{2} \cdot \frac{dx}{dt}$$

When the point is at $(0, -2)$, since the observer is turning at one radian per second, also $\frac{d\theta}{dt} = 1$. Also, looking at the diagram, $\theta = 0$. Plugging in these values:

$$\begin{aligned}\sec^2(0) \cdot (1) &= \frac{1}{2} \cdot \frac{dx}{dt} \\ 1 &= \frac{1}{2} \cdot \frac{dx}{dt} \\ \frac{dx}{dt} &= 2\end{aligned}$$

So, the particle is moving at 2 units per second.

(b) When the point is at $(0, 2)$, it is moving along a line with slope $-\frac{1}{2}$ and y -intercept 2. So, it is on the line

$$y = 2 - \frac{1}{2}x$$

That is, at time t , if the point is at $(x(t), y(t))$, then $x(t)$ and $y(t)$ satisfy $y(t) = 2 - \frac{1}{2}x(t)$. Implicitly differentiating with respect to t :

$$\frac{dy}{dt} = -\frac{1}{2} \cdot \frac{dx}{dt}$$

So, when $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = -\frac{1}{2}$. That is, its y -coordinate is decreasing at $\frac{1}{2}$ unit per second.

For the question “How fast is the point moving?”, remember that the velocity of an object can be found by differentiating (with respect to *time*) the equation that gives the position of the object. The complicating factors in this case are that (1) the position of our object is not given as a function of time, and (2) the position of our object is given in two dimensions (an x coordinate and a y coordinate), not one.

Remark: the solution below is actually pretty complicated. It is within your abilities to figure it out, but later on in your mathematical career you will learn an easier way, using vectors. For now, take this as a relatively tough exercise, and a motivation to keep learning: your intuition that there must be an easier way is well founded!

The point is moving along a straight line. So, to take care of complication (2), we can give its position as a point on the line. We can take the line as a sort of axis. We'll need to choose a point on the axis to be the “origin”: $(2, 1)$ is a convenient point. Let D be the point's (signed) distance along the “axis” from $(2, 1)$. When the point is a distance of one unit to the left of $(2, 1)$, we'll have $D = -1$, and when the point is a distance of one unit to the right of $(2, 1)$, we'll have $D = 1$. Then D changes with respect to time, and $\frac{dD}{dt}$ is the velocity of the point. Since we know $\frac{dx}{dt}$ and $\frac{dy}{dt}$, a reasonable plan is to find an equation relating x , y , and D , and differentiate implicitly with respect to t . (This implicit differentiation takes care of complication (1).) Using the Pythagorean Theorem:

$$D^2 = (x - 2)^2 + (y - 1)^2$$

Differentiating with respect to t :

$$2D \cdot \frac{dD}{dt} = 2(x - 2) \cdot \frac{dx}{dt} + 2(y - 1) \cdot \frac{dy}{dt}$$

We plug in $x = 0$, $y = 2$, $\frac{dx}{dt} = 1$, $\frac{dy}{dt} = -\frac{1}{2}$, and $D = -\sqrt{(0 - 2)^2 + (2 - 1)^2} = -\sqrt{5}$ (negative because the point is to the left of $(2, 1)$):

$$\begin{aligned} -2\sqrt{5} \cdot \frac{dD}{dt} &= 2(-2)(1) + 2(1) \left(-\frac{1}{2}\right) \\ \frac{dD}{dt} &= \frac{\sqrt{5}}{2} \text{ units per second} \end{aligned}$$

S-23: (a) Since the perimeter of the bottle is unchanged (you aren't stretching the plastic), it is always the same as the perimeter before it was smooshed, which is the circumference of a circle of radius 5, or $2\pi(5) = 10\pi$. So, using our approximation for the perimeter of an ellipse,

$$\begin{aligned} 10\pi &= \pi \left[3(a + b) - \sqrt{(a + 3b)(3a + b)} \right] \\ 10 &= 3(a + b) - \sqrt{(a + 3b)(3a + b)} \end{aligned}$$

(b) The area of the base of the bottle is πab (see Section A.10), and its height is 20 cm, so the volume of the bottle is

$$V = 20\pi ab$$

(c) As you smoosh the bottle, its volume decreases, so the water spills out. (If it turns out that the volume is increasing, then no water is spilling out—but life experience suggests, and our calculations verify, that this is not the case.) The water will spill out at a rate of $-\frac{dV}{dt}$ cubic centimetres per second, where V is the volume inside the bottle. We know something about a and $\frac{da}{dt}$, so a reasonable plan is to differentiate the equation from (b) (relating V and a) with respect to t .

Using the product rule, we differentiate the equation in (b) implicitly with respect to t and get

$$\frac{dV}{dt} = 20\pi \left(\frac{da}{dt}b + a\frac{db}{dt} \right)$$

So, we need to find the values of a , b , $\frac{da}{dt}$, and $\frac{db}{dt}$ at the moment when $a = 2b$.

The equation from (a) tells us $10 = 3(a + b) - \sqrt{(a + 3b)(3a + b)}$. So, when $a = 2b$,

$$10 = 3(2b + b) - \sqrt{(2b + 3b)(6b + b)}$$

$$10 = 9b - \sqrt{(5b)(7b)} = b(9 - \sqrt{35})$$

$$b = \frac{10}{9 - \sqrt{35}}$$

where we use the fact that b is a positive number, so $\sqrt{b^2} = |b| = b$.

Since $a = 2b$,

$$a = \frac{20}{9 - \sqrt{35}}$$

Now we know a and b at the moment when $a = 2b$. We still need to know $\frac{da}{dt}$ and $\frac{db}{dt}$ at that moment. Since $a = 5 + t$, always $\frac{da}{dt} = 1$. The equation from (a) relates a and b , so differentiating both sides with respect to t will give us an equation relating $\frac{da}{dt}$ and $\frac{db}{dt}$. When differentiating the portion with a square root, be careful not to forget the chain rule.

$$0 = 3 \left(\frac{da}{dt} + \frac{db}{dt} \right) - \frac{\left(\frac{da}{dt} + 3\frac{db}{dt} \right) (3a + b) + (a + 3b) \left(3\frac{da}{dt} + \frac{db}{dt} \right)}{2\sqrt{(a + 3b)(3a + b)}}$$

Since $\frac{da}{dt} = 1$:

$$0 = 3 \left(1 + \frac{db}{dt} \right) - \frac{\left(1 + 3\frac{db}{dt} \right) (3a + b) + (a + 3b) \left(3 + \frac{db}{dt} \right)}{2\sqrt{(a + 3b)(3a + b)}}$$

At this point, we could plug in the values we know for a and b at the moment when $a = 2b$. However, the algebra goes a little smoother if we start by plugging in $a = 2b$:

$$\begin{aligned}
 0 &= 3 \left(1 + \frac{db}{dt} \right) - \frac{\left(1 + 3\frac{db}{dt} \right) (7b) + (5b) \left(3 + \frac{db}{dt} \right)}{2\sqrt{(5b)(7b)}} \\
 0 &= 3 \left(1 + \frac{db}{dt} \right) - \frac{b \left(7 + 21\frac{db}{dt} + 15 + 5\frac{db}{dt} \right)}{2b\sqrt{35}} \\
 0 &= 3 \left(1 + \frac{db}{dt} \right) - \frac{22 + 26\frac{db}{dt}}{2\sqrt{35}} \\
 0 &= 3 + 3\frac{db}{dt} - \frac{11}{\sqrt{35}} - \frac{13}{\sqrt{35}} \frac{db}{dt} \\
 -3 + \frac{11}{\sqrt{35}} &= \left(3 - \frac{13}{\sqrt{35}} \right) \frac{db}{dt} \\
 \frac{db}{dt} &= \frac{-3 + \frac{11}{\sqrt{35}}}{3 - \frac{13}{\sqrt{35}}} = \frac{-3\sqrt{35} + 11}{3\sqrt{35} - 13}
 \end{aligned}$$

Now, we can calculate $\frac{dV}{dt}$ at the moment when $a = 2b$. We already found

$$\frac{dV}{dt} = 20\pi \left(\frac{da}{dt}b + a\frac{db}{dt} \right)$$

So, plugging in the values of a , b , $\frac{da}{dt}$, and $\frac{db}{dt}$ at the moment when $a = 2b$:

$$\begin{aligned}
 \frac{dV}{dt} &= 20\pi \left((1) \left(\frac{10}{9 - \sqrt{35}} \right) + \left(\frac{20}{9 - \sqrt{35}} \right) \left(\frac{-3\sqrt{35} + 11}{3\sqrt{35} - 13} \right) \right) \\
 &= \frac{200\pi}{9 - \sqrt{35}} \left(1 - 2 \left(\frac{3\sqrt{35} - 11}{3\sqrt{35} - 13} \right) \right) \\
 &\approx -375.4 \frac{\text{cm}^3}{\text{sec}}
 \end{aligned}$$

So the water is spilling out of the cup at about 375.4 cubic centimetres per second.

Remark: the algebra in this problem got a little nasty, but the method behind its solution is no more difficult than most of the problems in this section. One of the reasons why calculus is so widely taught in universities is to give you lots of practice with problem-solving: taking a big problem, breaking it into pieces you can manage, solving the pieces, and getting a solution.

A problem like this can sometimes derail people. Breaking it up into pieces isn't so hard, but when you actually do those pieces, you can get confused and forget why you are doing the calculations you're doing. If you find yourself in this situation, look back a few steps to remind yourself why you started the calculation you just did. It can also be helpful to write notes, like "We are trying to find $\frac{dV}{dt}$. We already know that $\frac{dV}{dt} = \dots$. We still need to find a , b , $\frac{da}{dt}$ and $\frac{db}{dt}$."

S-24: Since $A = 0$, the equation relating the variables tells us:

$$\begin{aligned}
 0 &= \log(C^2 + D^2 + 1) \\
 1 &= C^2 + D^2 + 1 \\
 0 &= C^2 + D^2 \\
 0 &= C = D
 \end{aligned}$$